# From Brains to Models: What AI Reveals About Being Human

Representation, understanding, and responsibility in medical AI

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**Abstract**

Artificial intelligence (AI) in medical imaging is both an engineering instrument and a comparative tool for thinking about cognition. A model may identify an image pattern, generate a useful explanation, or reproduce a human error without sharing the biological mechanism or lived experience associated with that performance. This educational perspective connects medical image analysis with questions about representation, attention, memory, embodiment, and consciousness. Marr's levels of explanation provide a framework for separating the purpose of a computation, the representations and algorithms that support it, and its physical implementation. A longitudinal brain-imaging vignette and a patient-facing report exercise connect this framework to clinical practice. Reproducible examples illustrate shortcut learning, prevalence-dependent predictive value, associative memory, and the differences between attention weights, gradients, and perturbation effects. They are deliberately synthetic and support conceptual learning rather than clinical claims. The central proposal is that comparisons between brains and models become informative when their scope, interventions, and failure criteria are explicit. Successful prediction can support useful action, but clinical reliance also requires evidence about context, uncertainty, workflow, and consequences. Embodiment and viability are treated as substantive hypotheses about cognition, not as established tests of consciousness or definitions of uniquely human value. The resulting educational task is to make both model capabilities and human responsibilities more precise.

**Keywords:** medical imaging; representation learning; cognitive neuroscience; clinical AI; consciousness; human oversight; AI education.

## Reader's guide

**Audience and starting knowledge.** For MSc and PhD students in biomedicine, imaging and computational science. You need percentages, graphs and the idea of a model fitted to data. Terms and notation are explained within the examples. Basic vectors and matrices help with the attention appendix; the core route assumes no differential equations or consciousness theory.

**Learning outcomes.** After working through the core examples, you should be able to:

1. Separate a task's purpose, its representation and algorithm, and its physical implementation.
2. Distinguish information available in a representation from evidence that a system uses it.
3. Explain how prevalence and the consequences of an action affect the interpretation of a prediction.
4. Distinguish a check of numerical implementation from evidence about biological validity or clinical benefit.
5. Propose a test of an alternative explanation and preserve uncertainty when communicating a result.

**Two reading paths through a shared core.** In Sections 1–7, follow the question, verbal example, figures and takeaway; then read the report exercise and synthesis (Sections 11–13). Both paths retain the assumptions, interpretations and check questions. The perspective boxes offer different starting points within selected technical examples. Choose either path and switch when the other helps; discipline does not determine mathematical preparation or clinical knowledge. Start with Notebook 00 for Marr's levels of explanation and Notebook 09 for a controlled imaging example.

- *Biomedical and imaging path:* use the MRI vignette (Section 3) to anchor each model claim. In growth modelling, distinguish the simulated density, the displayed boundary and an imaging observation (Section 4). In the vocabulary examples, identify the measured operation before making a brain comparison (Section 5). Use the expected counts to explain a positive result before considering an action (Section 7). The biomedical and imaging paragraphs in the boxes guide these comparisons.
- *Computational path:* after each verbal example, trace the variables, update rule and test. Follow the growth equation and numerical checks (Section 4; Notebooks 13 and 14a–14c), then the attention and Hopfield derivations (Appendices A and B; Notebooks 08, 10 and 12). Derive predictive value and the action threshold from counts and expected costs (Section 7; Notebook 15). The computational paragraphs guide these calculations; return to the shared interpretation to identify what the computed quantity measures.

**Optional extension for either path.** Reasoning/functional MRI studies, embodiment and consciousness (Sections 8–10), and clinical 4E examples (Appendix C), extend the same question: what observation would distinguish two explanations?

**Related resources:** Notebook guide and Colab links · Presentation guide. Each notebook runs independently. Predict, run, then explain the result.

# 1 Why this comparison belongs in a clinical AI school

A *medical image* is already a *mediated view* of a person. Acquisition hardware, pulse sequences or reconstruction algorithms, preprocessing decisions, and display conventions determine what can be seen. A *learned model* introduces another representation: a mask, probability, embedding, or sentence. A segmentation mask assigns a label to each image element; an embedding is a numerical vector used to represent an input. *Clinical interpretation* adds history and purpose. The patient then asks a question that none of these representations answers on its own: *what does this mean for my life?*

This sequence makes medical imaging a productive *meeting place for technical and philosophical inquiry*. It discourages both uncritical enthusiasm and a dismissal of models as mere calculation. Computation can be extraordinarily useful. The more important issue is which claim a computation warrants. A reproducible segmentation result is evidence about a specified image-analysis task. It is not automatically evidence about treatment benefit, biological mechanism, or the model's experience.

This educational perspective connects brain–model comparisons with foundational computational work, medical AI evaluation and consciousness research. It is a narrative synthesis rather than a systematic review. The examples distinguish published measurements, assumptions chosen for a model, and philosophical proposals; the synthetic calculations make those distinctions testable.

Deep learning enters multiple stages of magnetic resonance imaging (MRI), an imaging method using magnetic fields and radiofrequency signals to form tissue-dependent contrasts (Lundervold and Lundervold, 2019). Earlier expert systems made decision rules explicit; learned systems encode regularities in parameters fitted to data. This change alters the methods of development and inspection while preserving questions about evaluation, intended use, and responsibility.

**Takeaway.** An image, a model output and a clinical decision answer different questions.

**Check your understanding.** A segmentation becomes more accurate: which further observation would support a claim that it improves care?

# 2 Three levels of explanation, plus an explicit context

To compare a brain and a model, first ask which kind of explanation is needed. David Marr developed his framework in the study of vision, where a description of cells or computer instructions does not by itself explain how an image can support a description of the world. His book *Vision*, first published in 1982 and cited here in its 2010 edition, separates questions that are easily conflated (Marr, 2010). The motivation is explanatory: understanding a mechanism requires knowing the problem it solves, how it represents and transforms information, and how those operations are physically realized.

Consider estimating the volume of an enhancing lesion on MRI: a region that appears brighter on the relevant image after contrast administration. At the computational level, the task is to quantify the spatial extent of tissue meeting a specified imaging definition, so that examinations can be compared. At the representation-and-algorithm level, a binary mask assigns each voxel (a three-dimensional image element) to the target or background. Summing the target voxels and multiplying by voxel volume produces an estimate in millilitres. For example, 1,000 target voxels of 2 $mm^3$ each give $1{,}000 \times 2 = 2{,}000$ $mm^3 = 2$ mL. At the implementation level, the mask and summation are stored and executed on physical computing hardware. Knowing the memory layout does not tell us whether enhancement is the right target; knowing the intended comparison does not tell us how the mask was obtained. A clinician tracing the lesion and a network predicting its mask could support the same volume calculation through different

processes. This medical example separates agreement on an output from an explanation of the mechanisms producing it.

Marr distinguished a computational account of a task, a representation-and-algorithm account, and an implementation account (Marr, 2010). The framework is not a rigid pipeline or a ranking in which hardware is more fundamental than purpose. It *asks different questions about the same system.* A task can have multiple algorithms, and an algorithm can have multiple physical realizations. Conversely, the properties of an implementation can constrain which algorithms are feasible.

For segmentation, the computational question concerns what boundary or category should be estimated and why that output matters. The algorithmic question concerns the representation, architecture (the arrangement of computational components), loss (the function used to quantify training error), and inference procedure (how a fitted model produces an output). The implementation question concerns numerical precision, memory, devices, and execution. A U-Net's contracting and expanding paths explain aspects of its computational organization (Ronneberger et al., 2015); they do not by themselves specify the clinical meaning of its labels. The nnU-Net work further demonstrates how preprocessing, configuration, and training decisions belong to the effective method (Isensee et al., 2021).

Clinical purpose, environment, development and values influence what counts as a successful task. They form the context around Marr's three questions, rather than an additional level. The sections on embodiment and consciousness examine experience and viability—an organism's ability to remain alive—as further explanatory questions.

**Table 1:** Applying Marr's questions to the running example. The final row is contextual, not a fourth Marr level.

| Question | Medical imaging model | Brain and clinical reasoning |
|---|---|---|
| Purpose | Estimate a defined tissue label or response-related feature. | Interpret evidence to support an appropriate action. |
| Representation and algorithm | Features, masks, attention operations, losses, and decision rules. | Distributed representations, learning, memory, and inference; mechanisms require evidence. |
| Implementation | Devices, numerical operations, memory, and software. | Cells, synapses, neuromodulation, bodily systems, and development. |
| Context | Training population, workflow, incentives, and intended use. | Experience, relationships, bodily needs, and professional obligations. |

The framework also prevents a common rhetorical shortcut: counting parameters and neurons as though they were interchangeable units of intelligence. A parameter is a quantity in an engineered computation; a biological neuron is a dynamic cell within a complex physiological system. Morphological comparisons such as those reviewed by DeFelipe (2011) establish biological richness, not a conversion factor between brains and models. Similarly, the GPT-4 technical report does not disclose the model size, detailed architecture, hardware, or training compute (OpenAI, 2023). Comparisons should use disclosed or measured quantities with clearly defined system boundaries.

**Takeaway.** Marr's questions distinguish a purpose, a procedure and a physical realization while keeping their relationships visible.

**Check your understanding.** If a clinician and a network produce the same mask, what would you still need to learn about the algorithms producing it?

## 3 A running example: an image, a model, and a person

Consider a hypothetical multidisciplinary review of longitudinal brain MRI after treatment for glioma, a tumour of the central nervous system. An automated system segments enhancing tissue and produces a volume trajectory. A second model converts a technical radiology report into accessible language. A clinician must interpret both outputs in light of timing, symptoms, treatment history, and the patient's concerns. This vignette is not an account of an identifiable patient and contains no measured patient results.

Longitudinal datasets such as LUMIERE make repeated imaging and response-assessment research possible (Suter et al., 2022). Their existence does not make every downstream label unambiguous or every model externally valid. Enhancement is an imaging observation, whereas progression is a clinical interpretation made under specified criteria. Response Assessment in Neuro-Oncology (RANO) 2.0 illustrates the need for a structured response-assessment framework (Wen et al., 2023). The distinction to retain is that a changing image-derived quantity should not be silently promoted into a stronger clinical assertion.

An image-derived biomarker is a measured imaging characteristic used as an indicator of a biological process, disease process, or response to an intervention. Its meaning depends on a specified context of use (FDA-NIH Biomarker Working Group, 2016). Three questions must be separated. *Analytical validity* asks whether the measurement is sufficiently accurate and reproducible. *Clinical validity* asks whether it measures or predicts the clinically relevant process in the intended population. *Clinical utility* asks whether using it improves a decision or outcome. These are different evidential requirements: a repeatable number can be a poor indicator of progression, and a valid indicator need not improve care when acted upon.

In this vignette, enhancement reflects contrast behavior and cannot simply be equated with viable tumour. Treatment effects can complicate interpretation; timing, other imaging findings, and clinical status matter within response assessment (Wen et al., 2023). Acquisition differences, registration, and segmentation error can also change a measured volume. The inferential sequence is therefore: a measured change, a justified interpretation of its cause, and a decision about its consequences. Each transition needs evidence. An apparent increase should not become "the tumour is progressing" merely because a fluent report generator supplies that wording.

For a synthetic example, suppose enhancing volume changes from 10 to 12 mL. Compute the change relative to the starting volume:

$$\text{relative increase} = \frac{12 - 10}{10} \times 100\% = 20\%.$$

This is a statement about a measured quantity; measurement variability and cause are separate questions. If repeat measurements vary appreciably, the change may be difficult to distinguish from measurement error. Even a precisely measured increase does not identify its biological explanation. Finally, a change that supports concern about progression still needs to be considered alongside treatment options and the patient's goals. The values are illustrative; they are not a response threshold or a recommendation to alter treatment.

At each step, information is compressed. An image does not contain the entire person; a segmentation does not retain the entire image; a volume does not retain the segmentation's spatial arrangement. A short explanation may omit the uncertainty preserved in a longer report. Compression is essential for decision making, but its omissions should be appropriate to the decision. Figure 1 therefore includes two sources that cannot be replaced by image processing alone: clinical context and patient priorities.

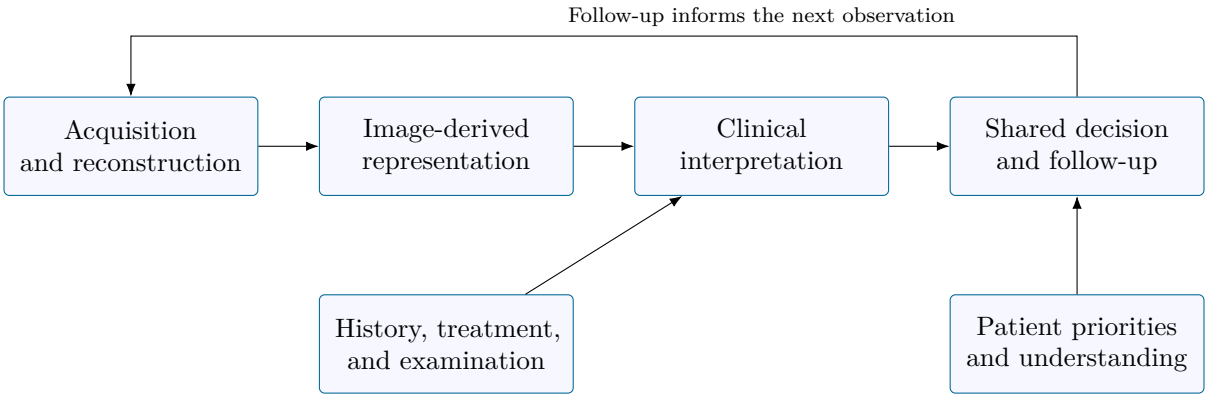



**Figure 1:** An original conceptual diagram of clinical interpretation. Arrows denote information or feedback, not validated causal effects. A model's output becomes useful within a larger process. The illustration synthesizes the imaging perspective of Lundervold and Lundervold (2019) and the clinical evaluation emphasis of Vasey et al. (2022).

### 3.1 A published glioma example: from compartments to regional burden

Lundervold et al. (2026) provide a concrete research counterpart to the hypothetical vignette. Their feasibility pipeline uses several complementary MRI contrasts: T1, gadolinium-enhanced T1, T2, and fluid-attenuated inversion recovery (FLAIR). T1 and T2 denote contrasts based on different tissue relaxation properties; FLAIR suppresses cerebrospinal-fluid signal. The pipeline combines these images with deep-learning tumour segmentation and subject-specific anatomical parcellation (division into named brain regions). It relates central non-enhancing/necrotic tumour, enhancing tumour, and surrounding oedema to brain regions, producing volumes and regional tumour-burden "hit-plots". Evaluation included 50 UCSF-PDGM glioblastoma cases and one longitudinal LUMIERE case with six timepoints. The reported agreement supports the evaluated segmentation task; the regional profiles supply anatomical context, while their effect on treatment decisions and outcomes remains to be established.

This example makes the distinction between representation and interpretation tangible: a tumour mask describes spatial membership, a volume summarizes extent, and a regional-burden profile describes overlap with named anatomy. Each summary preserves a different aspect of the image. Ask which aspect supports the intended clinical conclusion, and which evidence about tissue viability or function would have to come from another measurement.

Figures 2 and 3 reproduce source Figure A3 and Figure 6, respectively. The original legends retain source numbering; [60] denotes Suter et al. (2022). Detailed descriptions are added here.

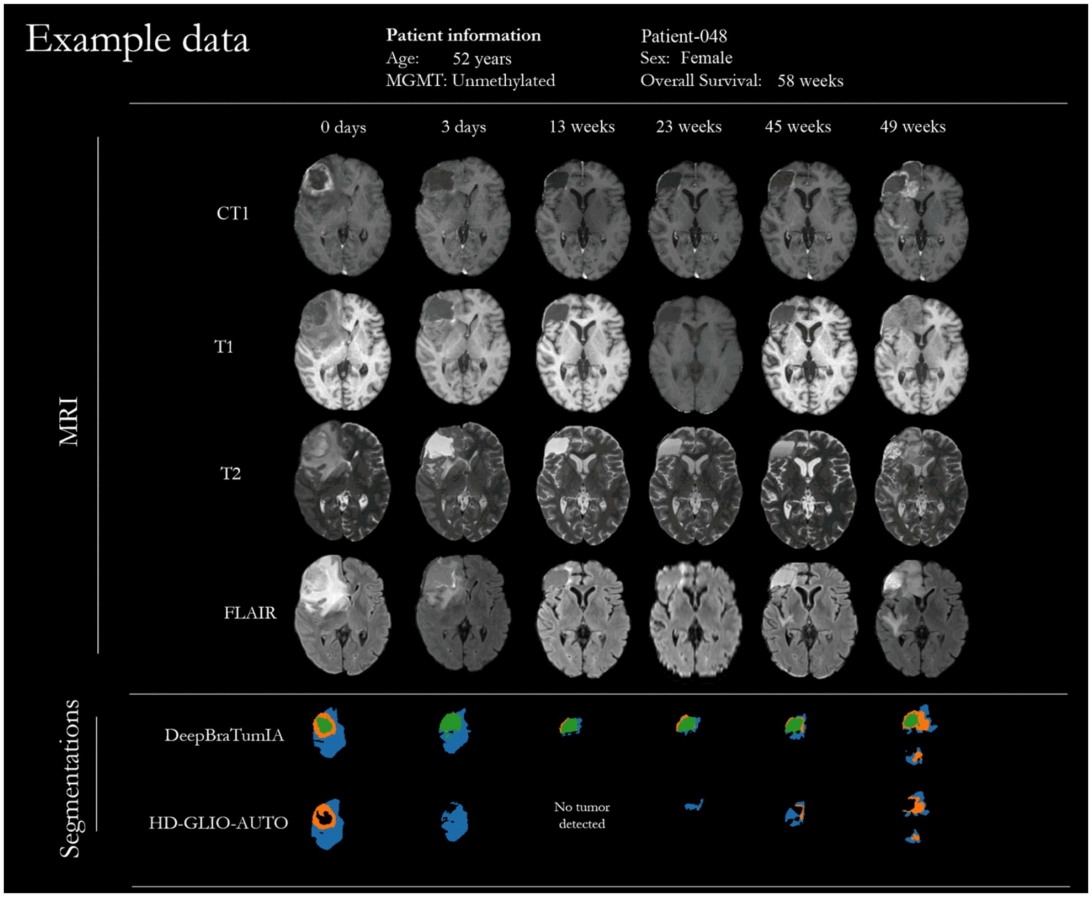



**Figure 2: Reproduction of Appendix Figure A3 from Lundervold et al. (2026). Original legend:** "Example of longitudinal data on GBM from the LUMIERE collection. CT1/T1: Post-/pre-contrast T1-weighted, T2-weighted, and fluid-attenuated inversion recovery (FLAIR). For this individual, the entire contrast enhancement was fully removed. Following a phase of complete response, there was a progression characterised by a highly rapid growth rate. In the present study we process all six mpMRI examinations of Patient-048 under the unified evaluation framework (Results, Figure 6). From Suter et al. [60] https://www.nature.com/articles/s41597-022-01881-7 (CC BY 4.0)."

**Detailed description.** Six columns follow the same published LUMIERE case from baseline and day 3 to weeks 13, 23, 45, and 49. Four axial MRI rows display contrast-enhanced T1 (CT1), unenhanced T1, T2, and FLAIR. The baseline images show a conspicuous lesion with enhancement and surrounding signal abnormality; the immediate postoperative images show the altered tumour bed. The later columns allow visual comparison of the low-enhancement interval and subsequent recurrence. Two lower rows show the corresponding colour-coded DeepBraTumIA and HD-GLIO-AUTO segmentation outputs. These are comparator outputs from the LUMIERE illustration, not the unified SegResNet output plotted next. Their colours should not be transferred to the next graph's legend. The "No tumor detected" annotation is an output of the displayed segmentation method, not proof of absence of all tumour cells. A single axial view illustrates morphology, while volumetry uses the three-dimensional masks. The source legend's description of complete removal should likewise be distinguished from the small postoperative enhancing volume estimated by the unified model.

**Attribution.** Image extracted unchanged from Appendix Figure A3, printed p. 28 (PDF p. 29), of Lundervold et al. (2026); the source credits Suter et al. (2022). Both articles identify CC BY 4.0 licensing. Image resized only; original legend retained, with typography reset. Licence: https://creativecommons.org/licenses/by/4.0/.

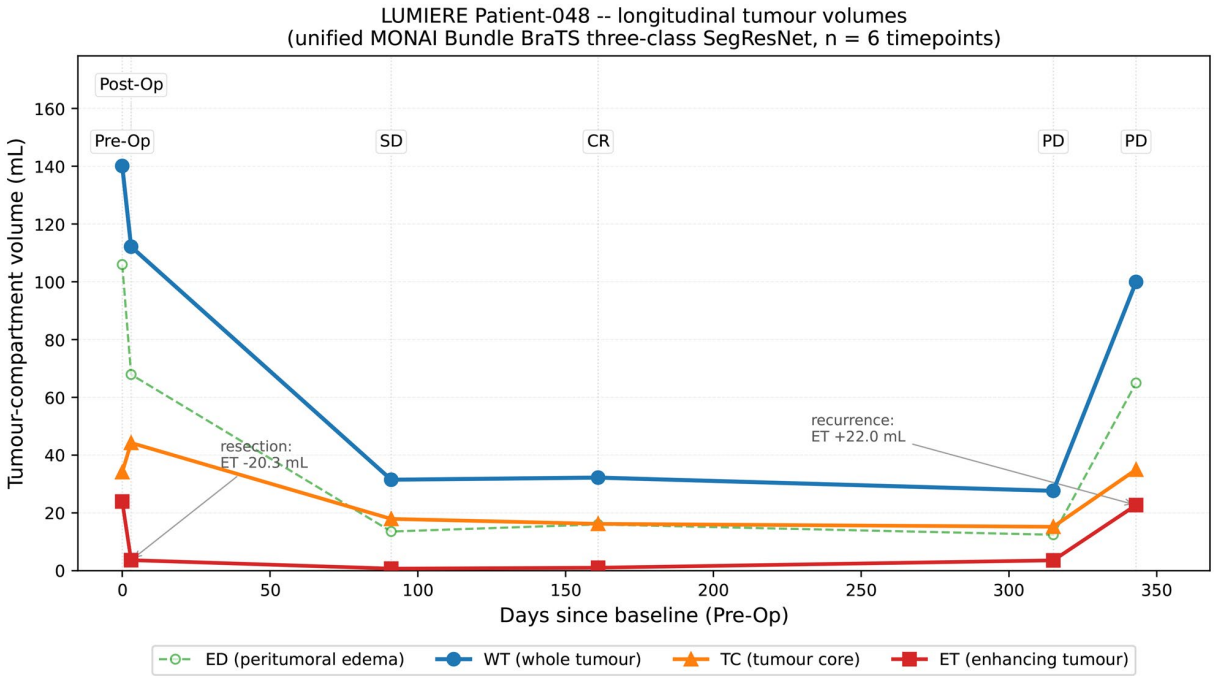



**Figure 3: Reproduction of Figure 6 from Lundervold et al. (2026). Original legend:** "LUMIERE Patient-048 – longitudinal tumour volumes from the unified MONAI Bundle BraTS three-compartment SegResNet ($n = 6$ timepoints). Per-compartment volume (mL) on a continuous day-resolved time axis: enhancing tumour (ET, red), necrotic core (NCR, blue), tumour core (TC=NCR+ET, dashed), edoema (ED, green), and whole tumour (WT=NCR+ED+ET, black). RANO labels (Pre-Op, Post-Op, SD, CR, PD, PD) and selected clinical event annotations (post-operative ET drop; recurrence trough/peak) are shown above the trajectories. LUMIERE anonymisation disclosure: the two baseline scans (week-000-1, week-000-2) carry day-resolution from the LUMIERE archive (0 and 3 days, respectively); from week-013 onward LUMIERE reports timepoints as floor-weeks [60], so days for week-013 / -023 / -045 / -049 are computed as week×7 and carry a ±3 day uncertainty."

**Verification note.** The published legend above does not match all plotted encodings. The image actually shows *four* series: WT as solid blue circles, TC as solid orange triangles, ET as solid red squares, and ED as dashed green open circles. NCR is not plotted separately. The image and original legend have been retained without correction; the description below follows the displayed plot.

**Detailed description.** The horizontal axis gives days since baseline; the vertical axis gives compartment volumes in millilitres. Six visits correspond to days 0, 3, 91, 161, 315, and 343, with the source's timing qualification retained above. WT falls markedly after the preoperative scan, remains much lower through the middle visits, then rises at the final visit. ET also drops sharply after surgery, stays near the volume floor, and increases again at recurrence. The source text reports WT/ET of 140.1/24.0 mL at baseline, postoperative ET of 3.7 mL, and final WT/ET of 100.0/22.7 mL. The displayed RANO labels identify stable disease (SD), complete response (CR), and progressive disease (PD), in addition to the surgical visits. These clinical categories are annotations, not classifications established by a volume curve alone. Lines connect discrete examinations; they are not continuous observations of tumour growth. TC and WT are composite volumes, so the four plotted quantities should not be summed.

**Attribution.** Image extracted unchanged from Figure 6, printed p. 15 (PDF p. 16), of Lundervold et al. (2026), © 2026 The Author(s), published by Informa UK Limited, trading as Taylor & Francis Group. CC BY 4.0: https://creativecommons.org/licenses/by/4.0/. Image resized only; original legend retained, with typography reset.

A useful exercise (a thought experiment) is to ask three people to state the output they need. The imaging scientist might require a reproducible boundary estimate. The clinician might require evidence that a change alters management. The patient might require an explanation of what remains uncertain and what happens next. These are compatible purposes, but they demand different evaluation targets. Confusing them makes technical improvements look more clinically complete than they are.

**Takeaway.** A measured volume change needs separate evidence about measurement reliability, biological interpretation and consequences for a decision.

**Check your understanding.** A volume increases from 10 to 12 mL: what would you need to determine whether the change reflects progression?

# 4 AI-assisted model development and numerical verification

Marr's questions help organize an AI-assisted simulation. The *task* is to investigate how cell spread and proliferation change a density field. The *representation* is density over space and time; the algorithm approximates the field's evolution. The *implementation* chooses a spatial grid, time steps, boundary treatment and numerical precision. Checking an implementation is distinct from testing whether the represented biology is adequate.

## 4.1 Read the growth equation

A population Fisher–Kolmogorov–Petrovsky–Piskunov (Fisher–KPP) model describes cell density $c(\mathbf{x}, t)$ at position $\mathbf{x}$ and time $t$ by

$$\partial_t c = \nabla\cdot(D\nabla c) + \rho c(1 - c/K), \qquad \mathbf{n}\cdot D\nabla c = 0 \quad \text{on the boundary.} \tag{1}$$

Read the equation as *change in density = redistribution through space + local growth.* Its quantities and operators have distinct roles:

- $\partial_t c$ is the rate of change at a fixed location. The symbol $\partial$ denotes a derivative with respect to one variable while the others are held fixed.
- $\nabla c$ is the spatial gradient: it records how density varies with position. The transport flux is $-D\nabla c$, toward lower density in isotropic diffusion. The divergence $\nabla\cdot(D\nabla c)$ gives the resulting local net gain or loss through transport.
- $D$, in $\mathrm{mm}^2/\mathrm{day}$, controls dispersal; it may be a scalar coefficient or a tensor describing directional differences. The proliferation rate $\rho$ is in $\mathrm{day}^{-1}$. Both $c$ and the carrying capacity $K$ are in $\mathrm{cells}/\mathrm{mm}^3$.
- The growth term is approximately $\rho c$ when $c$ is small compared with $K$, and becomes zero at $c = K$. The outward unit normal $\mathbf{n}$ specifies the direction perpendicular to the boundary. The boundary equation sets transport through that surface to zero.

The no-flux boundary specifies a closed transport domain; growth inside it can still change the total cell number. The migration/proliferation dichotomy known as *go-or-grow* is a separate mechanistic hypothesis (Hatzikirou et al., 2012); the displayed equation has no explicit switching between migratory and proliferative cell states.

**Two perspectives: growth modelling**

**Biomedical and imaging perspective.** Read $c$ as an assigned cell density and ask how a measurement would relate to it. In the threshold example below, first change only the display threshold at a fixed time, then change a growth parameter while keeping the threshold fixed. Which change alters the simulated state, and which changes only its displayed extent?
**Computational perspective.** Separate the evolution equation from the rule that converts its field into observations. Use Notebook 13 to check growth, transport and boundaries independently; use 14a–14c to investigate thresholding, anatomy and fitting. Track which assumptions are shared by the observation generator and fitted model before interpreting a withheld forecast.

## 4.2 Check the implementation using simpler cases

An AI assistant can propose equations, implement a solver and construct a display. To check the solver, remove transport by setting $D = 0$. At each location the equation becomes $dc/dt = \rho c(1 - c/K)$. For a starting density $c_0 > 0$ and constant $\rho, K$, its exact solution is

$$c(t) = \frac{K}{1 + (K/c_0 - 1)e^{-\rho t}}.$$

At $t = 0$ the denominator is $K/c_0$, giving $c(0) = c_0$; for $\rho > 0$, the solution approaches $K$ as time increases. Comparing the numerical trajectory with this formula tests the implemented growth term.

Next set $\rho = 0$ and retain zero boundary flux. Transport should redistribute cells while conserving their total number. A second check is needed for their location: periodic wraparound can conserve the total while moving cells across the wrong boundary. Reducing grid spacing and time steps tests how discretization error changes with resolution. Notebook 13 implements a small independent one-dimensional reference, plus a uniform-compartment coupled limit. An optional check exercises the pinned three-dimensional implementation on synthetic arrays without reading anatomy or patient data.

The separate BrainWeb demonstration adds an illustrative coupling of viable cells, oxygen and necrotic material, and compares it with a matched Fisher–KPP reference. Its reduced closure (the chosen relationships connecting the variables) and parameters are hypotheses for exploration. Removing oxygen uptake should recover the specified baseline; this checks consistency of the coupled equations. Testing physiological adequacy additionally requires independent oxygen and cell-density measurements, parameter estimation and withheld outcomes. The following exercises examine the simpler Fisher–KPP model on anatomy and with generated observations; they do not reproduce the full coupled implementation. The slide-to-resource guide records the external implementations and their versions.

## 4.3 Separate a density field from a displayed boundary

Notebook 14a extends verification into interpretation. It adapts the tissue and threshold experiments of the companion glioma teaching material (Lundervold, 2026). For a spatially constant carrying capacity, define the dimensionless fraction $u = c/K$. The equation becomes $\partial_t u = \nabla \cdot (D\nabla u) + \rho u(1 - u)$. An invented two-dimensional tissue map supplies a controlled comparison before introducing an anatomical phantom. Spatially varying cell motility in grey and white matter has a published glioma-model precedent (Swanson et al., 2000); the coefficients here are assigned teaching values, not estimates from that study or a particular MRI.

For pixel area $a$, two readings of the same field are

$$B_{2D} = a \sum_i u_i, \qquad A_\tau = a \sum_i \mathbf{1}[u_i \geq \tau].$$

The indicator $\mathbf{1}[\cdot]$ is one when the condition holds and zero otherwise. The integral $B_{2D}$ weights every pixel by its density fraction; $A_\tau$ counts selected pixels equally. Both have units $\text{mm}^2$, but neither is a cell count or whole-tumour volume. A 1 $\text{mm}^2$ pixel at $u = 0.8$ contributes 0.8 $\text{mm}^2$ to the integral and 1 $\text{mm}^2$ to $A_{0.2}$. Raising the reading threshold changes the displayed boundary without changing the simulated state.

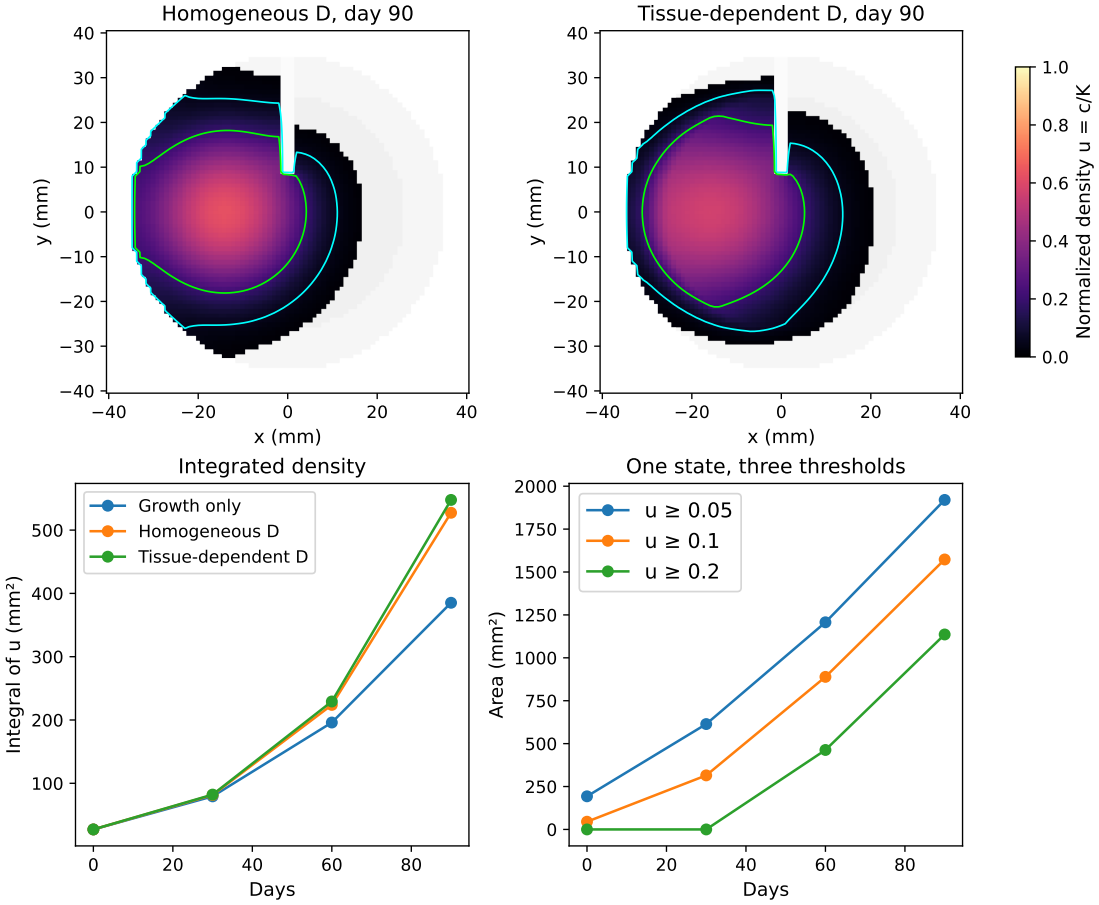



**Figure 4:** A constructed two-dimensional domain, not a patient image or BrainWeb slice. Homogeneous and tissue-dependent scalar diffusivities have the same domain mean, with the initial state and proliferation rate fixed. They need not produce the same spatial field or integrated density. The lower-right panel changes only the threshold applied to one simulation. The coefficients, closed barriers and thresholds are teaching assumptions.

## 4.4 Use anatomy without mistaking it for a tumour observation

Notebook 14b uses the original normal BrainWeb digital phantom and its tissue labels (Collins et al., 1998; Cocosco et al., 1997). Both input volumes are bundled with the teaching code; no companion checkout or separate data download is needed. Its simulated T1 contains no tumour. The density seed and growth rates are specified separately. The loader checks paired image geometry and file checksums, then samples every fourth native voxel at 4 mm spacing while retaining physical coordinates. This coarse representation can lose thin barriers; changing the sampling changes both the numerical grid and tissue geometry.

The growth domain contains grey matter, white matter and the phantom's glial class, which shares the grey-matter coefficient. CSF and other classes are excluded and act as no-flux barriers. We assign $D_{WM} = 0.5$ and $D_{GM} = 0.1$ $\text{mm}^2/\text{day}$ and $\rho = 0.04$ $\text{day}^{-1}$. A Gaussian seed with standard deviation 8 mm and peak $u = 0.12$ is centred at RAS+ coordinates $(-24, -12, 24)$ mm

and masked to that domain. RAS+ denotes axes increasing rightward, anteriorly and superiorly. A homogeneous comparison uses the mean coefficient over the same domain. These are spatially heterogeneous *scalar* coefficients, not fibre-directed diffusion tensors.

For voxel volume $v$ in mm$^3$, normalized burden is $B = v\sum_i u_i/1000$ in mL, while threshold volume is $V_\tau = v\sum_i \mathbf{1}[u_i \geq \tau]/1000$. At 4 mm spacing, one voxel is 0.064 mL: at $u = 0.25$ it contributes 0.016 mL to $B$ and 0.064 mL to $V_{0.2}$. In the specified tissue-dependent day-90 run, $B = 20.60$ mL, whereas $V_{0.05} = 93.12$ mL and $V_{0.2} = 36.67$ mL. These computed quantities are not observed lesion volumes. An absolute model-implied cell count would additionally require a justified $K$ in cells/mm$^3$; a comparison with MRI would require a measurement model for the relevant sequence.

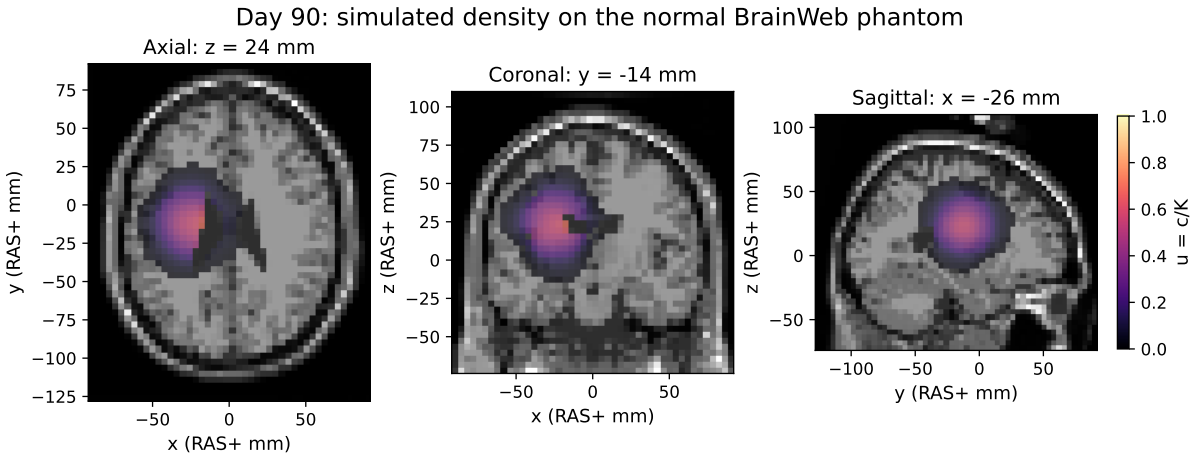



**Figure 5:** Simulated normalized density at day 90 on the normal BrainWeb phantom. Orthogonal planes pass through the specified seed, with actual sampled RAS+ coordinates labelled. The colour scale is fixed from zero to one; density below 0.01 is hidden for display and overlay opacity is 0.8. Geometry supplies anatomical context; the tumour field and rates are assigned. BrainWeb attribution: Collins et al. (1998); Cocosco et al. (1997).

## 4.5 Estimate parameters, then test a withheld forecast

A forward model computes a state from chosen parameters; an inverse problem estimates parameters from observations. Notebook 14c makes this distinction testable with generated observations on BrainWeb anatomy. The unknowns are a multiplier $s$ of both tissue diffusivities and a shared proliferation rate $\rho$. The white-to-grey ratio remains fixed at five. The generator uses the stated seed, $s = 1$ and $\rho = 0.04$ day$^{-1}$ on a 6 mm grid. Seed, tissue geometry, carrying-capacity normalization and ratio are known assumptions.

Define an observation operator $\mathcal{B}$ that averages each non-overlapping $2 \times 2 \times 2$ block into a 12 mm sample. Unmatched high-index edge cells are trimmed consistently. Independent Gaussian noise with standard deviation 0.005 in normalized-density units is added without clipping. Thus

$$y_i = \mathcal{B}[u(t_i; s, \rho)] + \epsilon_i.$$

This is a density-proxy measurement model, not an MRI simulator. A 72 mm cube around the seed is specified before noise generation and fitting. Least squares within that cube compares 63 candidate pairs using days 20, 40 and 60 only. Day 90 is withheld from parameter selection. The generator uses 0.5-day time steps and candidates use 1-day steps, but both share the spatial grid, equation and observation operator.

For noise seed 7, the grid search selects the generating pair, which is present among the candidates. At day 90 the RMSE (root mean squared error) against the withheld noisy spatial samples

is 0.00465, compared with 0.06288 for persistence, which simply carries the observed day-60 map forward. These are normalized-density units, not MRI intensity or clinical error scores. The search-grid minimum is not a confidence interval, and one favourable noise realization does not establish robustness or interval coverage. Unknown initialization, separate tissue rates and an imperfect observation operator would create additional inference questions. The source companion contains more extensive recovery and uncertainty experiments; those results are not claimed by this compact exercise.

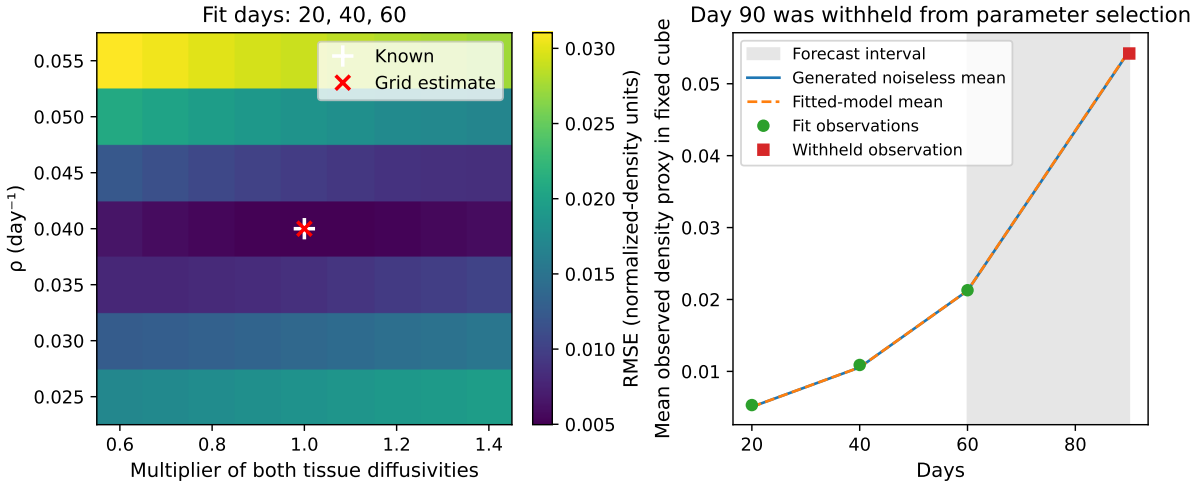



**Figure 6:** Generated-data recovery and a withheld forecast on the 6 mm BrainWeb grid. The left panel shows fitting error over the candidate grid. The right panel averages samples within the fixed observation cube for display; fitting and reported RMSE use the spatial samples individually. Day 90 does not enter fitting. Shared model and measurement assumptions make this a controlled inference exercise, not independent biological validation.

**Interactive exploration.** Notebooks 14a–14c provide live controls for tissue rates, numerical settings and inference choices. Time, threshold and image-view controls in 14a/14b read cached states; changing simulation parameters requires a new run. Notebook 14c varies noise, early observation times and search resolution while keeping day 90 withheld. The paper's figures retain the fixed reference settings stated above.

This example gives domain expertise a concrete role in AI-assisted research: specify the scientific question, check units and assumptions, compare with independent solutions, inspect failures, and distinguish an implemented proposal from an empirically supported model.

**Takeaway.** A limiting case can check a solver against an independent answer; biological adequacy needs measurements of the system being modeled.

**Check your understanding.** A diffusion solver conserves total cells but wraps transport around the boundary: which additional test would reveal the error?

# 5 Shared vocabulary needs operational definitions

## 5.1 Representation: similar geometry is informative, but incomplete

A representation makes some distinctions available to a downstream process: a later computation that uses it. For a model, this can mean that a layer's activations support a classifier or predict neural responses on held-out stimuli. For a person, representations are inferred from behavior, physiology, and interventions. Their relationships to meaning, action, and experience require

additional arguments. "Representation" is useful precisely because it is broad, but that breadth can conceal changes in the claim being made.

For a concrete model example, freeze an image encoder (keep fixed the weights that map an image to a feature vector) and fit a linear classifier to its embeddings to distinguish lesion-containing from lesion-free patches. Evaluate it on patients excluded from both encoder training and classifier fitting. Successful decoding indicates that this distinction is available to that readout under those conditions. It does not show that the encoder's original output head uses it, or that the signal is the lesion rather than a correlated feature.

For a human comparison, ask participants to categorize objects viewed from unfamiliar angles and test whether an independently fitted encoding model predicts their measured visual responses. Behaviour, neural prediction, and a causal intervention address different claims. Recognizing a cup across views, reaching to grasp it, and experiencing its appearance are related phenomena, but a decoder of object category measures only one part of that relationship. An intervention that disrupts grasp selection would add evidence about action; category decoding alone would not establish the experience of seeing.

Yamins et al. (2014) tested a more specific connection. They optimized hierarchical models for object categorization and assessed their ability to predict neural responses to images in macaque visual cortex. The stimuli varied object appearance and included background variation. Higher model representations predicted inferior temporal responses, while intermediate representations predicted V4 responses. The important point is that an engineering objective helped select representations informative about independently measured biology; a network was not simply assigned a cortical label because its diagram looked hierarchical. This is a substantive bridge between engineering and neuroscience. Its evidential force depends on the stimuli, recording conditions, model comparisons, and evaluation procedure. Predicting some neural responses does not identify the biological learning rule. It also does not show that every layer has a one-to-one cortical counterpart. A model may approximate an input-output relation while implementing it differently.

An optimization-based approach to neuroscience asks how objective functions, learning rules, and architectures jointly constrain behavior (Richards et al., 2019). For medical AI, this motivates comparing models under changed conditions rather than only comparing average test scores. Does a representation retain the lesion information needed after an acquisition change? Does it support appropriate abstention—declining to provide an ordinary prediction—when a required sequence is absent? These questions connect representation to its practical role without assuming that a successful embedding explains the whole person.

An illustrative experiment makes the optimization argument tangible. Train the same architecture with either a lesion-label objective or an image-reconstruction objective; separately hold the objective fixed while changing the architecture or learning rule. Use matched data and report any unmatched training budget. Then test whether lesion information remains decodable and usable after a controlled change in image contrast. The first comparison examines the effect of the objective; the others examine architectural and learning constraints. This is a proposed experiment, not a result attributed to Richards et al. (2019).

In the MRI vignette, compare lesion masks on paired images with identical anatomy but a specified synthetic intensity transformation, and then validate on an independent acquisition setting. Report lesion-wise failures and boundary changes as well as average overlap. Separately remove a required sequence and record whether a system designed to abstain does so. Preserved average performance, preserved lesion information, and appropriate abstention are three distinct outcomes. Synthetic transformations help isolate a question, but do not reproduce every consequence of a scanner change.

**Takeaway.** Information decodable from a representation is available to the chosen readout; an intervention tests whether a particular output uses it.

**Check your understanding.** A lesion label is decodable from an embedding: what intervention would test whether the original classifier depends on lesion information?

## 5.2 Attention: a weighting operation and a family of human capacities

A Transformer is a neural-network architecture that updates representations using attention and other operations. Its attention operation lets one input position combine information from other positions using weights that depend on their representations. The positions can contain image patches or tokens: units such as words, word parts or characters. A query specifies what is being matched, keys provide the quantities against which it is matched, and values carry the information to be combined. For example, an image-patch representation can be updated using information from other patches. These are learned numerical transformations, rather than literal questions or labels supplied by a person. Appendix A gives the equation, dimensions, and a worked calculation (Vaswani et al., 2017).

This computational definition does not define human attention, which is studied through selective processing, task performance, and neural mechanisms. The shared name invites hypotheses about selection; it does not establish identity. A useful teaching question is: what information is being selected, by which operation, and for which output?

A heatmap is a display format, not a single explanation method. An attention map displays internal weighting coefficients. A gradient saliency map displays derivatives: how a chosen output changes under sufficiently small input changes. A perturbation map compares outputs after specified input changes. These quantities answer different questions; none becomes a clinical explanation simply by being overlaid on an anatomical image.

To see the distinction, imagine a classifier whose overlay highlights an enhancing region. This may be compatible with appropriate use of lesion information, but visual plausibility alone cannot exclude dependence on another feature. A saliency method can even produce plausible structure after learned parameters have been randomized. Model-parameter and training-label randomization are therefore useful sanity checks: an explanation that claims to describe a learned decision should be examined for dependence on that learning (Adebayo et al., 2018). Passing these checks is a starting point, not proof of a biological mechanism.

A complementary exercise is to change a suspected acquisition cue while preserving the lesion as far as possible, then inspect the prediction and explanation together. Such experiments also need controls: masking a region can create an unfamiliar image, so a changed prediction may reflect the intervention artifact. The explanatory claim and the validity of its test both require scrutiny. When an interpretable model can meet the clinical task, the trade-off should be examined explicitly (Rudin, 2019).

For a specific controlled example, put a bright site marker outside the anatomy in otherwise identical synthetic images. Change only that marker and recompute the score and maps. The anatomy is exactly preserved by construction, so a score change establishes sensitivity to the marker in this model. In a real exported image, remove or replace an existing border annotation without altering anatomical pixels, and compare with an equally sized control edit elsewhere in the border. The latter comparison helps detect sensitivity to the editing procedure. It still cannot establish that the marker explains all errors or that arbitrary inpainting is a valid counterfactual.

Appendix A.3 and `code/src/brainsmodels/attention_maps.py` make these distinctions executable. A small, explicitly specified self-attention sublayer and scalar readout produce an attention matrix, a signed input gradient, and a patch-occlusion map for the same synthetic input (notebook 08; the two-key calculation is in notebook 10). The example also changes a

marker while holding the central pattern fixed and randomizes parameters. It is a transparent computational exercise, not a trained clinical Transformer or a reproduction of a saliency validation study.

**Two perspectives: attention and image overlays**

**Biomedical and imaging perspective.** Before interpreting an overlay, identify what its colours encode and which output is being examined. In the synthetic marker experiment, compare the score before and after the marker changes while the central pattern stays fixed. Anatomical plausibility and dependence of a prediction on a feature are separate questions.
**Computational perspective.** Work through Appendix A with Notebook 10: keep query and keys fixed, then change the second value from 10 to $-10$. Explain why the weights stay fixed while the output changes. In Notebook 08, compare a local derivative with a finite replacement, keeping the output target and replacement baseline explicit.

**Takeaway.** Attention weights, gradients and finite replacements describe different operations and must be interpreted with their output and scale.

**Check your understanding.** If attention weights stay fixed while the values change, must the final weighted output stay fixed?

### 5.3 Memory: distinguish parameters, context, retrieval, and recollection

A language model estimates probabilities for tokens given a preceding context. In a deployed language-model system, "memory" can refer to trained parameters, the current context (input available for the current calculation), an external retrieval store, or an application-maintained record. These differ in persistence, update mechanism, provenance, and access. A statement available through retrieval may disappear when the index changes; a statement encoded in parameters may be difficult to locate or revise. Neither should be casually equated with an autobiographical episode.

For example, trained parameters may support completing a familiar phrase; a current prompt may contain a newly supplied fictional identifier; retrieval may supply a versioned teaching protocol; and an application may reload the preference "explain abbreviations." Only the last three examples require access to those particular records at inference time. Even there, the mechanisms can overlap: a retrieved passage usually becomes part of the generator's context.

Autobiographical remembering adds a different explanatory target. Remembering one's first day in a laboratory can include a place, an event sequence, personal significance, and a sense of having been there. A system could store and accurately reproduce a dated account of that day, but this establishes access to information about an event, not the first-person experience of recollecting it. The distinction does not prove that artificial recollection is impossible; it specifies why factual accuracy alone cannot settle the question.

Consider a report assistant that correctly recalls a patient's stated preference within a conversation. The preference may be present in the current input, loaded from an application record, or retrieved from a document. Correct output alone does not reveal which mechanism supplied it. Removing the earlier message, disabling retrieval, or changing the stored record are different interventions. They help identify the source of the behavior without assuming that the system has recollected an event.

Retrieval-augmented generation explicitly combines parametric and external information (Lewis et al., 2020). Access to a source does not guarantee that the source was selected correctly or used faithfully. In a clinical report exercise, one should therefore record both the retrieved passage and the generated statement. Provenance is a property to inspect, rather than infer from the confidence of the language.

**Table 2:** Four engineering uses of "memory." Persistence depends on the deployed system; these are mechanisms, not human memory categories.

| Mechanism | Concrete illustrative example | A discriminating check |
|---|---|---|
| Parameters | A familiar phrase can be completed without providing its source in the prompt; learned weights influence the completion. | Compare documented model versions or controlled training interventions. |
| Context | A prompt states that fictional sample Q7 has label blue; the next answer uses that statement. | Remove the relevant passage while holding the other input fixed. |
| Retrieval | A versioned teaching protocol is retrieved from an index and supplied with the question. | Inspect retrieved sources and repeat with retrieval disabled. |
| Application record | Software reloads the saved preference "explain abbreviations" at the start of a new session. | Inspect what was saved and what is actually reintroduced. |

In complementary learning systems theory, rapid hippocampal learning and slower neocortical learning help reconcile two needs: acquiring a particular event and integrating regularities across events without excessive interference (McClelland et al., 1995). A teaching example is learning a new colleague's name after one introduction while gradually learning the conventions of a new laboratory. These are illustrative cases, not clean anatomical assignments: the theory concerns interacting systems, and learning rate alone does not identify a brain region.

An engineering comparison could store each new meeting note immediately in an external index while updating a general model only periodically. This separates rapid item availability from slower changes in generalization. However, retrieval into a fixed-weight generator does not itself implement hippocampal–neocortical consolidation. To test the narrower analogy, add conflicting records and then revise selected facts. Measure item-specific recall, interference with older items, and transfer to questions whose answers are not copied from any one record. If apparent "integration" disappears whenever retrieval is disabled, the claim that knowledge was consolidated into model parameters is unsupported. Conversely, persistent performance alone is insufficient if the answer was already learned; novel controlled facts and held-out questions are needed.

Why introduce a Hopfield network here? It isolates a modest but recognizable operation: recovering a whole pattern from a corrupted cue. Think of recognizing a familiar pattern despite missing or misleading details. In the model, "remembering" means that repeated local updates settle into a stored configuration. This makes associative retrieval explicit enough to inspect mathematically (Hopfield, 1982); it does not model the feeling of recollection or the history of a personal episode.

Its didactic advantage is that storage and retrieval become different operations that students can inspect. During storage, pairwise weights are set from a pattern; during retrieval, those weights remain fixed while the state changes. A partial cue can therefore be completed without an exact address lookup or further weight training. Students can corrupt a cue, predict which unit will change next, and then check both the recovered pattern and the energy. This connects the earlier distinctions between parameters, transient state, and retrieval to an executable mechanism.

Imagine recognizing a familiar melody after hearing only fragments. Pattern completion is the selected functional analogy; the 12 binary units below do not represent a realistic auditory circuit. The model's value is to generate questions: how much corruption can retrieval tolerate, which competing pattern is recovered, and can the dynamics settle confidently into the wrong state? A wrong attractor is especially useful pedagogically because it separates convergence from correctness.

Figure 7 shows a 12-unit example in which three signs of a stored pattern have been flipped. Each unit is updated in turn using a weighted signal from the others. A scalar energy tracks the settling process. Under the stated assumptions, each actual change lowers the energy, but an unchanged update leaves it flat. Appendix B derives this result; notebook 12 records each update. Successful settling is not guaranteed to recover the intended memory in more complex cases: there can be competing or spurious attractors.

Can this tell us how the brain might work? It demonstrates the sufficiency of one recurrent mechanism for content-addressable pattern completion and suggests tests of attractor-based accounts. A biological prediction is that degrading retrieval cues should expose a dependence on circuitry that supports completion. In mice, Nakazawa et al. (2002) found impaired recall under reduced spatial cues after a CA3-targeted NMDA-receptor manipulation, despite preserved performance under full cues. This supports a role for CA3-related processes in associative recall; it does not establish the Hopfield update equation or isolate all contributions of encoding and retrieval.

Stronger tests would compare neural population trajectories and behavioural errors across cue quality, competing memories, and interventions, against alternative models such as feed-forward classification. The classical network assumes symmetric weights, binary states, and asynchronous updates, whereas biological circuits have richer dynamics and plasticity. Its energy is a mathematical function, not a measurement of metabolic expenditure. Thus the model can reveal a candidate computational principle and its failure modes while leaving its biological realization open.

**Two perspectives: associative retrieval**

**Biomedical and imaging perspective.** Compare the corrupted cue, final pattern and energy trace in Notebook 12. Ask separately whether the system settles and whether it retrieves the intended pattern. Then identify the behavioural and neural observations that would be needed to investigate pattern completion in a biological system.

**Computational perspective.** Follow one asynchronous update in Appendix B, checking where symmetry and the zero diagonal enter the energy argument. Next start at the sign-reversed target: it is stable and has the same energy as the target in this construction. Report retrieval accuracy alongside energy; a decreasing energy trace alone does not select the intended memory.

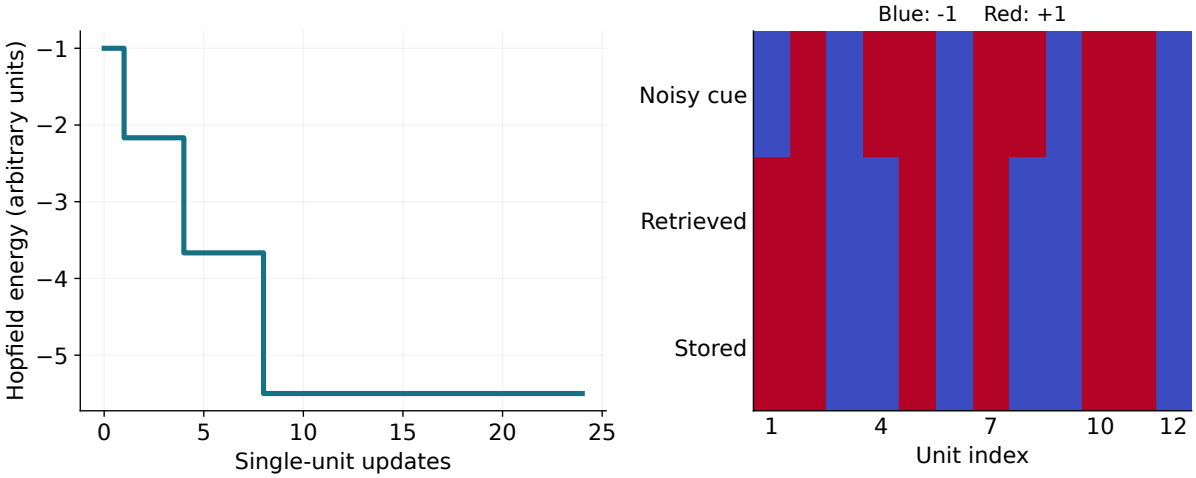



**Figure 7:** A computed 12-unit Hopfield example with one stored pattern and three flipped bits in the initial cue. The left panel records the energy after individual asynchronous updates; the right panel compares states. The result illustrates associative retrieval under specified assumptions, not human recollection or consciousness. Source code and numeric outputs accompany the paper.

**Table 3:** A vocabulary for careful brain–model comparison. Each row identifies a question rather than a verdict on equivalence.

| Term | An operational model meaning | What remains to be established |
|---|---|---|
| Representation | Information usable by a downstream computation. | Which biological or semantic relationships the representation captures. |
| Attention | Input-dependent weighting in a specified architecture. | Relationship to human selection, control, and awareness. |
| Memory | Parameters, context, retrieval, or stored application state. | Relationship to learning, consolidation, recollection, and selfhood. |
| Reasoning | Performance on inference tasks with explicit tests. | Robustness, mechanism, causal competence, and transfer. |
| Understanding | A stipulated capacity, such as explanation and counterfactual use. | Whether performance supports the intended stronger interpretation. |

**Takeaway.** Parameters, transient context, external retrieval and application records have different update and persistence mechanisms; pattern completion makes one retrieval mechanism explicit.

**Check your understanding.** A report assistant reproduces a saved preference: which controlled removal would help identify where the preference was supplied?

# 6 Compare systems through controlled tests

A brain–model comparison becomes testable when it names an ability and asks which mechanisms could produce it. Searching for a uniquely human remainder by subtracting model successes can obscure this question. If a model succeeds at a task, the result demonstrates that the implemented system is sufficient for that tested behavior under those conditions. It does not establish that its components are necessary for the behavior, or that humans use the same components. If it fails, the cause might be a poor interface, insufficient training, a mismatched task, or a genuine missing capacity.

A stronger protocol therefore starts with a named ability and competing explanations. For example, both a clinician and an image classifier may detect a lesion. One explanation is sensitivity to relevant anatomy; another is reliance on a correlated acquisition cue. Change the cue while preserving the target as far as technically possible, evaluate the change in behavior, and check whether the intervention itself introduces artifacts. Repeat across alternative models and appropriate human comparison groups. Control training exposure and test contamination where feasible, and acknowledge when they cannot be matched.

Agreement in aggregate accuracy is particularly weak evidence for shared mechanisms. Two systems may succeed on different cases, express different uncertainty, or fail for different reasons. Compare error structure, responses to perturbations, and the information available to each system. Ablation can establish dependence on a component within a particular implementation, but the component may be replaceable in a different implementation. These distinctions are also relevant to claims about human-like learning (Lake et al., 2017).

## 6.1 A transparent shortcut experiment

Suppose a binary label $Y$ is balanced (half the cases have each label) and a site marker $S$ agrees with it with probability $a$. A shortcut is a predictive cue that works in the development data but need not remain reliable under the intended conditions of use. A fixed rule $\widehat{Y} = S$ has expected

accuracy $a$. Suppose another binary image feature agrees with the label with probability 0.8, independently of the site–label relation. A fixed feature rule then has expected accuracy 0.8 under the stipulated changes. At $a = 0.95$, the site rule wins. At $a = 0.50$, it falls to chance. At $a = 0.05$, its predictions are usually wrong. Figure 8 follows directly from these assumptions; no classifier has been trained and no hospital data have been analyzed.

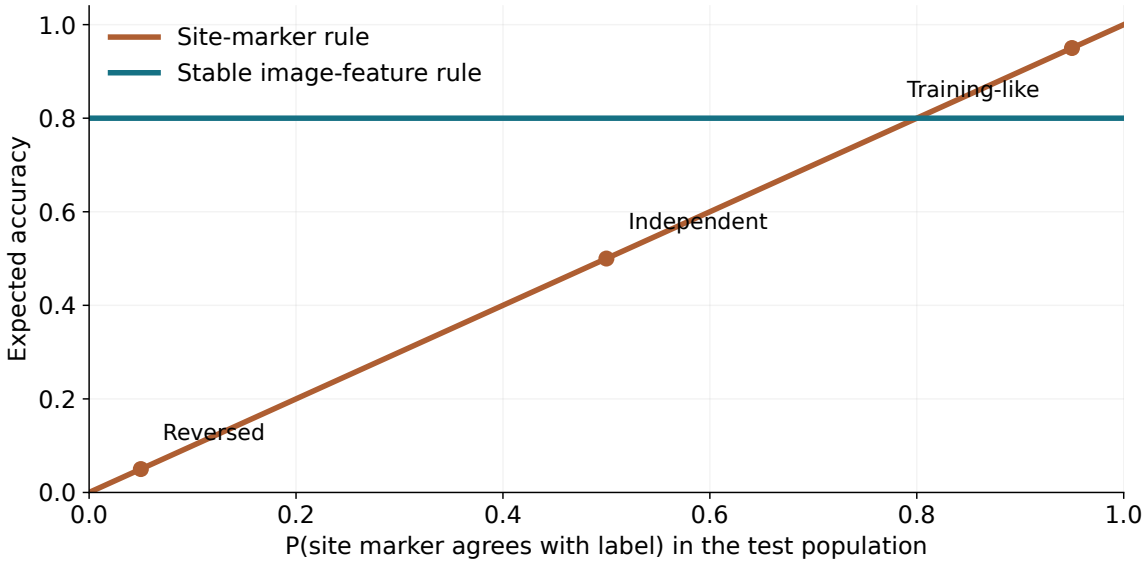



**Figure 8:** Analytic teaching example of shortcut dependence. The site rule's accuracy equals the marker–label agreement probability. The stable feature rule remains at 80% by construction. The figure illustrates how a ranking can reverse across environments; it does not estimate the performance of a medical AI system.

The broader shortcut-learning problem is documented across deep learning (Geirhos et al., 2020). In chest radiography, cross-hospital evaluation has exposed variable generalization and hospital-associated signals (Zech et al., 2018). These findings motivate tests tailored to the proposed shortcut. Keeping each patient entirely within one data partition helps prevent leakage (evaluation information entering development). Site-aware and temporal evaluation test transfer across institutions and time; inspecting acquisition cues tests a proposed source of dependence. A random split alone may leave important correlations untouched. Notebook 11 reconstructs the three analytic accuracy comparisons.

**Takeaway.** A useful comparison changes a suspected cause while controlling alternatives; average accuracy alone leaves many mechanisms compatible with the result.

**Check your understanding.** The site rule falls from 95% to 5% accuracy: which stipulated change produces the reversal?

## 7 Prediction acquires meaning inside a decision

A probability is useful when its target and intended use are clear. *Ranking* concerns the order of cases: do cases with the condition tend to receive higher scores? *Calibration* concerns numerical probabilities: among comparable cases assigned a probability near 0.2, does the condition occur about 20% of the time? *Utility* concerns the consequences of using the output. A model can rank cases well while assigning overly extreme probabilities. Calibration methods can improve probability estimates on suitable evaluation data (Guo et al., 2017); calibration should also be checked in the intended population, with uncertainty and subgroup sample size reported.

### 7.1 Count the positive results before using the formula

For a specified target condition and classification threshold, *sensitivity* is the fraction of condition-present cases classified positive. *Specificity* is the fraction of condition-absent cases classified negative. *Prevalence* is the fraction of the population with the condition. *Positive predictive value* (PPV) asks the reverse question from sensitivity: among positive results, what fraction have the condition?

Consider 10,000 hypothetical people with 1% prevalence, 90% sensitivity and 90% specificity. First, $0.01 \times 10{,}000 = 100$ have the condition and 9,900 do not. Next, sensitivity gives $0.9 \times 100 = 90$ true positives. The false-positive fraction is $1 - 0.9 = 0.1$, giving $0.1 \times 9{,}900 = 990$ false positives.

**Table 4:** Expected counts in the hypothetical 10,000-person example. Rows group people by condition status; columns group them by the classifier's result.

| Condition status | Positive result | Negative result | Total |
|---|---|---|---|
| Present | 90 | 10 | 100 |
| Absent | 990 | 8,910 | 9,900 |
| Total | 1,080 | 8,920 | 10,000 |

There are $90 + 990 = 1{,}080$ positive results, so PPV $= 90/1{,}080 \approx 8.3\%$. The low prevalence supplies many more opportunities for false positives than true positives. This explains why "90% accurate" cannot substitute for a specified performance measure and population.

The same counting argument gives Bayes' formula. In a population of size $N$, the expected true-positive count is $N\,\mathrm{Se}\,\pi$ and the expected false-positive count is $N(1 - \mathrm{Sp})(1 - \pi)$, where $\pi$ is prevalence, Se sensitivity and Sp specificity. Dividing true positives by all positives cancels $N$:

$$\mathrm{PPV} = \frac{\mathrm{Se}\,\pi}{\mathrm{Se}\,\pi + (1 - \mathrm{Sp})(1 - \pi)}. \tag{2}$$

Keeping sensitivity and specificity at 90%, PPV is 50% at 10% prevalence and 90% at 50% prevalence (Figure 9). These are hypothetical calculations. The fixed sensitivity and specificity assumption isolates the effect of prevalence; case mix and acquisition can also change those quantities in practice.

### 7.2 Compare the expected consequences of acting

Let $p$ be the probability of the target condition given the available evidence, used as the decision probability in this simplified model. Let $C_{\mathrm{FP}}$ be the cost of acting when the condition is absent and $C_{\mathrm{FN}}$ the cost of not acting when it is present. If correct actions have zero cost, the expected costs are

$$L_{\mathrm{act}} = (1 - p)C_{\mathrm{FP}}, \qquad L_{\mathrm{no\ act}} = pC_{\mathrm{FN}}.$$

Choose the action when its expected cost is lower. For positive costs, rearrange $(1-p)C_{\mathrm{FP}} < pC_{\mathrm{FN}}$ to obtain

$$p > \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}. \tag{3}$$

For an arithmetic example, if a missed condition costs four units and an unnecessary action costs one, the threshold is $1/(1 + 4) = 0.2$. These stipulated costs demonstrate the calculation. A clinical decision model would also need justified estimates of treatment harms, delays, competing outcomes, repeated tests and patient preferences. The mathematical point is that an objective

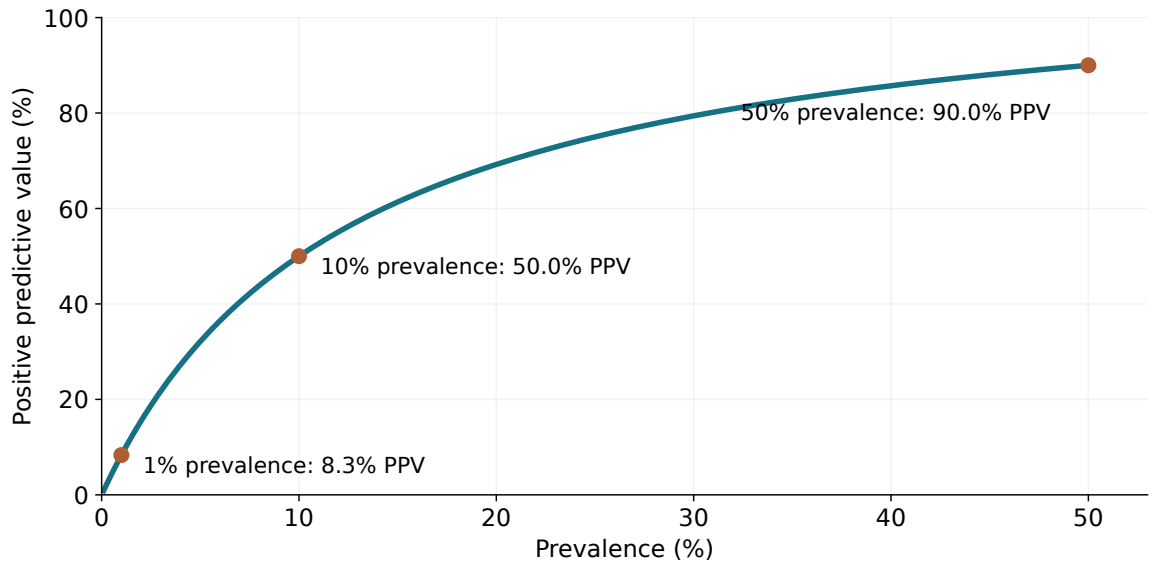



**Figure 9:** Positive predictive value under fixed 90% sensitivity and 90% specificity. The plotted values are calculated from Bayes' rule. Changing prevalence changes the meaning of a positive classification even before considering calibration drift or altered test performance.

does not specify its own acceptable consequences. Someone must justify the target, the actions, and the values represented by their costs.

Notebook 15 reproduces the counts and cost comparisons, with separate controls for population rates, decision probability and costs.

**Two perspectives: predictive value and decisions**

**Biomedical and imaging perspective.** Use the 10,000-person table to explain why only 90 of 1,080 positive results are true positives in this example. Identify the population to which those counts refer. Before proposing an action, ask which consequences and patient preferences the simplified cost model leaves out.

**Computational perspective.** Reconstruct the expected counts and cancel the population size to obtain PPV. Keep population prevalence $\pi$ distinct from the decision probability $p$. Derive the action threshold by comparing the two expected costs, then ask whether the probabilities are calibrated in the population where the rule will be used.

## 7.3 Evaluate the system people actually use

A workflow may change who sees a case, when they see it, and what they believe before reviewing the image. A nominal human checkpoint does not establish effective oversight. The reviewer needs relevant information, sufficient time, the ability to challenge an output, and an actionable fallback. Performance should be assessed for the combined human–AI process, including errors introduced or amplified by that process.

Clinical studies can investigate this process directly. The MASAI randomized mammography screening study illustrates evaluation of an AI-supported reading workflow rather than a model in isolation (Lång et al., 2023). Its findings should be interpreted within the intervention and endpoints studied, not generalized to every imaging modality. Reporting guidance provides complementary structure: STARD-AI addresses diagnostic accuracy studies, DECIDE-AI early live clinical evaluation, and TRIPOD-LLM studies involving large language models (Sounderajah et al., 2025; Vasey et al., 2022; Gallifant et al., 2025). A completed checklist is not evidence of benefit; it makes the design and evidence easier to assess.

**Table 5:** Moving from a technical result to justified clinical reliance. These are examples of evidence questions, not a certification checklist.

| Claim | Evidence to examine | What it does not establish alone |
|---|---|---|
| A mask is accurate | Relevant reference labels; overlap and boundary errors; small-lesion failures. | Correct diagnosis, response assessment, or treatment benefit. |
| A score transports | External, temporal, and subgroup evaluation; calibration and missing-input behavior. | Utility at the intended action threshold. |
| A report is clearer | Fidelity, omissions, uncertainty preservation, and reader comprehension. | Appropriate advice or improved outcomes. |
| Oversight works | Observed review behavior, correction, workload, escalation, and fallback. | That a nominal human checkpoint prevents harm. |
| The system helps | Prospective workflow comparison and outcomes appropriate to intended use. | Durability after model, population, or workflow changes. |

**Takeaway.** A positive result depends on prevalence and test performance; an action also depends on justified consequences and the workflow.

**Check your understanding.** In the 10,000-person example, what happens to false positives and PPV if specificity rises while prevalence and sensitivity stay fixed?

# 8 Research extension: observing reasoning and brain representations

The core examples have distinguished an output from the mechanism producing it. This extension applies the same question to language-model reasoning and brain imaging. It examines representations and algorithms in Marr's framework: what can be observed, what can be decoded from that observation, and what further test would show that a candidate representation contributes to an output?

## 8.1 Model interfaces and the observability of reasoning

An interface specification describes available inputs, outputs and resource limits of a large language model (LLM) application. A long context window specifies how much input may be available to a computation; it does not establish reliable recall of every item or autobiographical memory. Reproducible experiments should record the model version, application configuration, prompts and tools separately from claims about internal architecture or measured performance.

A chain of thought is a sequence of generated intermediate reasoning steps, usually expressed in text. Monitoring it raises a different question: can an observable reasoning trace help detect a specified property of the computation or resulting action? A trace may supply useful evidence, but it is not a complete record of internal activity. Research on monitorability emphasizes that relevant computation may remain unexpressed, and that changes in training or reasoning methods can weaken this source of evidence (Korbak et al., 2025). A displayed explanation or summary should also be distinguished from the underlying trace available to an evaluator.

The underlying computational ideas have primary research examples. *Coconut* abbreviates *Chain of Continuous Thought*: it is a method for reasoning through numerical hidden states rather than emitting a natural-language token at every intermediate step (Hao et al., 2024). Its last-layer hidden state—a vector of activations—is fed back as the next input embedding. Training progressively replaces explicit reasoning steps with such latent steps; ordinary token

generation can resume to produce an answer. This is a particular model-and-training procedure, not a name for every recurrent network.

Recurrent-depth models instead apply an internal block repeatedly, allocating additional computation before producing an output token (Geiping et al., 2025). A schematic description is

$$h^{(k+1)} = F_\theta\Big(h^{(k)}, x\Big), \qquad k = 0, \ldots, K-1. \tag{4}$$

Here $x$ denotes the encoded task input or conditioning context; $h^{(k)} \in \mathbb{R}^d$ is the numerical latent state after internal step $k$; $d$ is its dimension; $F_\theta$ is the update operation with learned parameters $\theta$; and $K$ is the number of internal updates. The superscript $(k)$ indexes an iteration, not a power. The initial state $h^{(0)}$ is supplied by the model's initialization or input-processing stage. The parameters are held fixed during this schematic inference loop. An actual transformer may maintain a sequence of vectors and cached context, so the equation suppresses implementation details. By contrast, an external text loop decodes and appends tokens between steps. Both can supply additional inference-time computation; neither establishes a human-readable inner language or identifies the architecture of a proprietary system.

## 8.2 From fMRI representations to interpretation and understanding

Functional MRI (fMRI) and language-model monitoring both require an observation to be interpreted through a model, but they expose different kinds of evidence. The blood-oxygen-level-dependent (BOLD) fMRI signal measures haemodynamic changes related to neural activity, rather than individual neural spikes (Logothetis et al., 2001). Its spatial pooling and delayed temporal response constrain inferences about rapid processing. A chain-of-thought trace is generated symbolic output; an activation probe reads numerical states when these are accessible. Neither channel becomes an explanation simply because an investigator can decode information from it.

Three operations should be distinguished. An *encoding model* predicts measured brain responses from stimulus features, for example contextual language-model embeddings. A *decoder* predicts stimulus content or a report from brain measurements. An *explanatory test* asks whether a proposed feature or mechanism survives a discriminating experiment. For encoding, independent test stories or images are essential, with temporal alignment and feature selection fitted without test leakage. For decoding, comparisons with language-only, image-only, shuffled-input, and zero-input controls establish what the brain signal adds. For explanation, new stimuli or targeted interventions must challenge alternatives. Similar representational geometry is evidence of a relationship under a chosen analysis, not proof of identical computation.

A foundational example is Tang et al. (2023): their fMRI decoder generated language capturing aspects of perceived and imagined speech and silent-video content. Its dependence on participant cooperation illustrates why successful decoding in a trained setting should not be generalized to unrestricted access to thought. More recent work extends the comparison in several directions. The following studies illustrate different tests of representation and interpretation.

**Visual meaning in a language-model space.** Doerig et al. (2025) showed that embeddings of scene captions predict fMRI responses to viewed scenes and support caption reconstruction. Controlled comparisons implicated contextual information beyond individual words; image networks trained to predict these embeddings also acquired representations aligned with brain responses. This provides a quantitative account of some high-level visual information, while leaving open whether humans use the model's representational format or learning procedure.

**Readable hypotheses tested in the scanner.** Antonello et al. (2026) developed *generative causal testing*: descriptions of what drives a brain-response model are used to generate new linguistic stimuli, and the resulting brain responses test those descriptions. The method connects prediction to experimentally tested selectivity. Manipulating stimulus content strengthens an explanation of response tuning, but does not by itself identify a necessary neural circuit or explain subjective understanding.

**Sparse features and competing predictors.** In the cited preprint, Lepori et al. (2026) replaced dense language-model states with sparse-autoencoder features (a learned representation that uses relatively few active components) and included surprisal as a separate predictor. Surprisal is $-\log P(w \mid \text{context})$: an unlikely word $w$ has greater surprisal. Using 7-T fMRI from eight participants listening to 200 sentences, they identified interpretable response profiles, including sensitivity to people-related content; frontal responses were relatively well explained by surprisal alone. This is a useful bridge between model interpretation and brain encoding, and a reminder to test simpler predictors before attributing explanatory power to a complex embedding.

**Inner speech and the contribution of context.** The *MindAlign* preprint maps fMRI during silent image description into a semantic space (a vector representation of meaning-related distinctions), then combines that representation with visual context to prompt a frozen multimodal language model (Liu et al., 2026). The authors report improvements over fMRI-only and random baselines. Because the image supplies information about the intended description, the central explanatory question is what neural input adds beyond an image-only account. The trained task should not be equated with unconstrained inner-speech reading.

**A recent decoding failure that sharpens the test.** The preprint by Suvakovic et al. (2026) reports that apparently strong candidate-ranking performance in an fMRI-to-language system remained essentially unchanged when fMRI inputs were zeroed. The result is specific to their system and evaluation; it does not invalidate neural decoding generally. It does show why fluent output and above-chance ranking cannot establish brain-signal use without appropriate controls. Their decoding regimes and test stories also differ across comparisons, limiting broad performance conclusions.

These studies suggest a concrete connection to monitoring LLM inference. A probe can reveal that information is available in a hidden state, just as a decoder can recover information from fMRI; demonstrating that the system *uses* that information requires further evidence. In an accessible model, activation replacement, ablation, or a controlled input change can test whether the candidate representation contributes to an output. In a person, experimentally varied stimuli, behavioural measurements, and appropriately justified neural interventions answer related but non-identical questions. A readable reasoning trace, an activation-based interpretation, and an fMRI reconstruction should therefore be evaluated against a specified target, such as detecting an error or distinguishing two hypotheses, rather than treated as interchangeable windows into thought. Understanding advances when the proposed account predicts what will change under a new test.

**Takeaway.** Decoding information and demonstrating its contribution to an output require different tests.

**Check your understanding.** If candidate-ranking performance is unchanged after fMRI inputs are zeroed, what contribution of the brain signal has that evaluation established?

## 9 Embodiment and viability: a hypothesis with limits

A useful introduction is the family of *4E approaches*: cognition is considered as embodied, embedded, enactive, and extended (Newen et al., 2018). Embodied emphasizes bodily contributions; embedded emphasizes environmental and social supports; enactive emphasizes cognition through active engagement; extended asks whether some tightly integrated external resources constitute parts of a cognitive process. These approaches overlap, but they are not one theory with a single test. Appendix C develops the distinctions through the imaging vignette.

For this paper, their value is to broaden the unit of analysis. A clinician's interpretation depends on a body, a working environment, active inspection, and external records. Whether an external resource merely supports cognition or partly constitutes it is a further philosophical claim (Clark and Chalmers, 1998). Workflow dependence can be studied without first deciding that stronger question.

A substantive philosophical proposal concerns prediction under viability constraints: the conditions an organism must maintain to remain alive. Biological cognition occurs in organisms that regulate bodily conditions, develop over time, act in an environment, and can be injured. This gives perception and action a practical significance that is not captured by a task score alone. Predictive and interoceptive accounts (concerning sensing and regulation of internal bodily conditions) make aspects of this proposal computationally explicit (Friston, 2010; Seth, 2013).

The proposal should not be inflated into the claim that all behavior is governed by a single literal survival objective. People pursue social, intellectual, moral, and self-destructive goals. Nor is embodiment uniquely human: it characterizes many living systems. Calling it the human remainder would therefore confuse a contrast with certain engineered systems and a definition of humanity. The defensible claim is that bodily regulation and lived context deserve explicit consideration when interpreting human cognition and clinical needs.

The artificial side needs equal care. A base language model's next-token training objective is not a complete description of every deployed AI system. Post-training, multimodal input, retrieval, external tools, memory, and robotic interaction alter the system being evaluated. Some artificial systems act through bodies and feedback loops. Being embodied in an engineering sense does not itself settle whether a system has biological needs or experience. Conversely, the absence of biological metabolism is not a proof that experience is impossible.

A more precise comparison therefore names the systems: for example, an adult human regulating bodily state and a specified language-model application with fixed weights, a given context window, and defined tools. It asks which feedback signals are available, what changes through learning, how goals are represented, and what counts as failure. These questions can be investigated without first settling a universal boundary between humans and machines.

**Takeaway.** Body, environment and feedback can change the appropriate unit of analysis; a proposed role should be tied to an observable comparison.

**Check your understanding.** How would you distinguish an effect of active image exploration from an effect of seeing different images?

## 10 Consciousness: experience, access and evidence

"Consciousness" can name different explanatory targets. Here, *phenomenal consciousness* means subjective experience: there being something it is like to see a colour, feel pain, or hear a sound. *Access consciousness* concerns information being available for use in reasoning, report, and deliberate control (Block, 1995). The first asks about experience; the second asks about availability for cognitive use. These are conceptual distinctions, not two established anatomical

compartments.

Consider seeing a red warning on a display. Its visual appearance illustrates the phenomenal question. Using the warning to explain a concern or decide what to inspect next illustrates the access question. The example distinguishes questions about an ordinary event; it does not demonstrate that the two properties can be cleanly separated experimentally. Wakefulness, selective attention, self-awareness, and verbal report are also relevant concepts, but none should be used as a synonym for all consciousness.

Global neuronal workspace theory proposes that information becomes widely available through interactions among specialized processors. Such broadcasting is a proposed mechanism for aspects of conscious access; it is not simply a restatement of the definition. Integrated information theory makes different commitments about the causal organization relevant to experience (Seth and Bayne, 2022). A theory, a measured correlate, and a definition must therefore be kept distinct when interpreting experiments.

An AI system can produce a sentence about pain, uncertainty, or self-awareness. The sentence is an observable output. Whether it provides evidence of experience depends on additional assumptions about the system and the relation between report and consciousness. The same caution applies to the reverse inference: inability to give a verbal report does not show absence of experience. This matters in medicine, where patients may have impaired communication.

An adversarial collaboration, in which proponents of competing theories agree on tests, examined predictions of global neuronal workspace and integrated information theories (Cogitate Consortium et al., 2025). It does not establish a universally accepted test for consciousness in artificial systems. Theory-derived indicator approaches offer a structured way to assess AI while retaining uncertainty about both the theories and their application (Butlin et al., 2026).

Two positions deserve a fair hearing. A functionalist approach holds that relevant organization could, in principle, be implemented in more than one substrate. A biologically oriented approach argues that details of living organization may matter in ways current models omit. Neither position is resolved by a fluent conversation or by showing that a model uses matrix multiplication. The purpose of the keynote is to explain why the question remains open and how stronger evidence might be sought, not to announce that consciousness has been detected or excluded.

There is also an ethical distinction between determining a machine's possible moral status and determining its effect on people. A nonconscious system could profoundly alter a patient's care, expectations, or opportunities. Clinical responsibility therefore cannot wait for agreement about machine experience. People and institutions remain the parties who select, deploy, oversee, and respond to the consequences of clinical systems.

**Takeaway.** Experience, access, report and proposed mechanisms are different explanatory targets.

**Check your understanding.** A model accurately reports information from an earlier input: which observation concerns access, and what additional argument would be needed to infer experience?

## 11 A report exercise that bridges into RadKom

The RadKom workshop on patient-friendly radiology language provides a practical setting for this exercise. Consider the invented sentence: *"The enhancement is indeterminate; treatment-related change and progression remain possible."* Ask participants to produce a plain-language rendering while preserving both possibilities and the uncertainty.

A reasonable answer is: *"The scan shows a change whose cause is not yet clear. It may be related*

*to treatment, or it may mean the tumour has grown."* The source supplies two possible causes and no definite plan. The rendering preserves those possibilities without adding a diagnosis, treatment recommendation or promised follow-up.

A technically fluent model may make the passage shorter and warmer while changing its meaning. An expert may preserve meaning while using language that is hard to understand. These are separate dimensions and should be evaluated separately. In Marr's terms, preserving clinically relevant meaning is the task, the wording and transformation procedure are the representation and algorithm, and the particular model/application supplies the implementation. Record the input, exact model or system version, prompt, output, and human revisions. Reviewers assess contradiction, unsupported additions, omissions, uncertainty, and readability. Patient or public involvement should shape what counts as understandable communication in the intended setting.

This example also connects to a linguistic debate. Bender and Koller distinguish linguistic form from meaning and caution against unexamined claims of understanding (Bender and Koller, 2020). Their argument concerns the relationship between a learning setup and a stronger semantic claim; it should not be turned into a blanket empirical verdict on all multimodal or interactive AI. For this exercise, the operational question is more tractable: did the transformation preserve the source's clinically relevant propositions and their uncertainty?

**Takeaway.** Plain language should preserve the source propositions and their uncertainty while making them easier to understand.

**Check your understanding.** Does replacing "may mean the tumour has grown" with "the tumour has grown" preserve the source meaning?

## 12 Teaching transferable judgment

A mixed cohort benefits when each technical concept ends in an interpretable question. After the attention example, ask what operation it implements; use Appendix A for the calculation. After the Hopfield example, ask which assumptions ensure energy decreases. After the shortcut plot, ask which test would distinguish useful anatomy from acquisition cues. After the prevalence calculation, ask what additional information is needed before acting. The same activity can be approached computationally by one participant and conceptually by another.

Use the perspective boxes for a paired explanation: one reader explains what a result could mean for an observation or decision, and another reconstructs the calculation and its assumptions. Exchange roles before proposing a further test. Both readers should be able to state what was computed and which interpretation the evidence supports.

The educational aim is not simply to teach students how to obtain an answer from a model. It is to make them specify what they would accept as evidence and notice when an answer exceeds that evidence. A practical assessment can ask students to submit a result together with one falsification attempt: a changed population, a perturbed input, a contradictory example, or a check of the source text. An unsuccessful model can still teach something if the failure is characterized accurately; a successful one can still be misunderstood.

The accompanying code uses small deterministic examples that run on an ordinary CPU. No large-model download is required. Keeping the demonstrations modest supports inspection and reproducibility. For substantive model experiments, resource reporting should name the model, precision, context, software, workload, and measurement method. Memory estimates for weights alone exclude runtime buffers and other overhead.

A useful collaboration between the two school communities would pair a computational claim with a conceptual audit. One participant states the target, data, and test. Another asks what the target leaves out, whose purposes it serves, and which stronger interpretations might

be mistakenly inferred. Both then revise the experiment. This is not an opposition between engineering and ethics; the questions jointly determine whether the technical result is relevant to the proposed use.

**Takeaway.** An informative exercise reports both a result and an attempt to challenge its interpretation.

**Check your understanding.** Choose one notebook result: which alternative explanation would your next controlled test address?

## 13 Conclusion

AI can serve as a comparative instrument for studying human abilities and as a practical instrument for medicine. Its value in both roles increases when the claims become narrower and more testable. Shared outputs do not by themselves imply shared mechanisms. Shared terminology does not settle shared experience. A successful model needs a justified connection to the actions people take with it.

The most useful conclusion for clinical integration is therefore a disciplined allocation of questions. Ask the model what it can estimate under specified conditions. Ask the experiment which alternative explanations have been excluded. Ask the clinical team and patient what matters for the decision. Being human need not be defined as whatever the latest model cannot yet do. In medicine, our responsibilities become clearer when we understand both the power of these models and the lives into which their outputs enter.

**Takeaway.** Specify the computation, test competing explanations, and connect the result to the purposes of the people using it.

**Check your understanding.** For a model claim you encounter next, what single observation would make you revise your interpretation?

## Reproducibility and provenance

The training examples and simulated tumour states are generated for teaching; the glioma and fMRI studies are empirical research identified separately. Reproduced figures contain published LUMIERE case information; no patient imaging volumes or additional patient records are distributed here. The normal BrainWeb T1 phantom and tissue labels used in Notebooks 14b/14c are bundled with the code, with source attribution and checksums. The presentation guide maps topics to local notebooks and versioned external implementations. The notebook guide supplies learning goals, setup instructions and Colab links. AI assistance was used in preparing text and code; scientific interpretation and responsibility rest with the named author. Source, dependencies and reproduction commands are recorded in https://github.com/arvidl/brains-to-models. A short repository note documents the preserved background manuscript.

## A Attention: from similarity scores to a weighted output

### A.1 The operation and its dimensions

For one attention head, let $X \in \mathbb{R}^{n \times d_{\text{model}}}$ contain $n$ token or image-patch representations, one per row. The notation $\mathbb{R}^{n \times d}$ means a matrix of real numbers with $n$ rows and $d$ columns; $d_{\text{model}}$ is the number of features per representation. Learned projection matrices give $Q = XW_Q$, $K = XW_K$, and $V = XW_V$, with $Q, K \in \mathbb{R}^{n \times d_k}$ and $V \in \mathbb{R}^{n \times d_v}$. Thus $d_k$ is the query/key

dimension and $d_v$ the value dimension. A superscript $\top$ denotes transpose, exchanging rows and columns. Scaled dot-product attention is

$$A = \mathrm{softmax}_{\mathrm{row}}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right), \qquad Z = AV. \tag{5}$$

The $n \times n$ matrix $A$ contains normalized weights; $Z$ has shape $n \times d_v$. For scores $s_{ij}$, row-wise softmax computes $A_{ij} = e^{s_{ij}} / \sum_\ell e^{s_{i\ell}}$: exponentiate each score and divide by the total in its row. An optional mask $M$ excludes disallowed positions by adding $-\infty$ to their scores, making their exponential zero; allowed positions receive zero in $M$. With finite allowed scores and at least one allowed key, each row of $A$ sums to one. Causal masking prevents access to later positions when predicting a sequence one token at a time. In cross-attention, queries and keys/values come from different sequences and $A$ need not be square (Vaswani et al., 2017).

## A.2 An original two-key calculation

Take one query $q = (1, 0)$, two keys $k_1 = (1, 0)$ and $k_2 = (0, 1)$, and scalar values $v_1 = 2$, $v_2 = 10$. With $d_k = 2$ and no mask, the scores are $z_1 = 1/\sqrt{2}$ and $z_2 = 0$. Thus

$$\alpha_1 = \frac{e^{1/\sqrt{2}}}{e^{1/\sqrt{2}} + 1} \approx 0.6698, \qquad \alpha_2 \approx 0.3302, \tag{6}$$

$$o = \alpha_1 2 + \alpha_2 10 \approx 4.6419. \tag{7}$$

The larger match receives more weight, but the other value still contributes. If both scores were equal, the output would be their average, 6. If only the first key were allowed by the mask, the output would be 2. These are consequences of this toy calculation, not measurements from a trained model.

The weights alone do not determine the output. Keeping the same query and keys while changing $v_2$ from 10 to $-10$ leaves the displayed attention weights unchanged, yet changes the output from about 4.6419 to $-1.9629$. This is a simple counterexample to treating a weighting plot as a complete explanation of the computation. The example is reproduced in `code/src/brainsmodels/clinical_appendix.py`.

To check the intermediate arithmetic, $e^{1/\sqrt{2}} \approx 2.0281$, so the weights are approximately $2.0281/3.0281$ and $1/3.0281$. Using the unrounded weights, their contributions to the original output are $\alpha_1 v_1 \approx 1.3395$ and $\alpha_2 v_2 \approx 3.3024$. The second value contributes more despite its smaller weight. In the intervention, only the sign of that second contribution changes. Notebook 10 performs the calculation at full precision.

Why divide by $\sqrt{d_k}$? Under the illustrative assumptions that query and key components are independent, zero-mean, and unit-variance, their dot product has variance $d_k$. Scaling gives variance one. Large unscaled scores can concentrate softmax weights and reduce its local sensitivity. This calculation motivates the scale; it does not assert that trained representations meet those distributional assumptions exactly (Vaswani et al., 2017).

Multiple heads apply different projections and their outputs are combined. A full Transformer also has other operations, including feed-forward transformations and residual paths. Accordingly, an attention matrix describes one intermediate operation. It neither identifies human attentional mechanisms nor by itself establishes which input features causally determine the final prediction.

## A.3 Executable maps from one self-attention sublayer

The accompanying NumPy exercise uses 16 synthetic image patches and one summary token (CLS). Each row of $X \in \mathbb{R}^{17\times 4}$ contains an intensity, two position coordinates, and a CLS

indicator. The projection matrices are explicitly specified in the code. They are not fitted to data. The sublayer and scalar output are

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \qquad Y = X + AVW_O, \qquad f(X) = Y_{0,:}w.$$

Here $W_Q, W_K \in \mathbb{R}^{4\times 2}$, $W_V, W_O \in \mathbb{R}^{4\times 4}$, and $w \in \mathbb{R}^4$. Attention uses Equation 5 without a mask. The residual path is retained, but layer normalization, a feed-forward block, multiple heads, and training are omitted so that every dependency is inspectable. This is the attention operation used within a Transformer, not a complete trained Transformer. The output $f$ is an arbitrary score, not a clinical probability.

Three maps are computed for this same output. The attention map is the CLS row $A_{0,1:17}$ reshaped onto the patch grid; the CLS self-weight is omitted without renormalization. The signed gradient is $\partial f/\partial X_{j,0}$: it measures local sensitivity to patch intensity while holding position fixed. The occlusion effect is $f(X) - f(X^{(j\leftarrow b)})$, where only patch $j$'s intensity is replaced by baseline $b$. A positive gradient means a sufficiently small intensity increase raises the score. A positive occlusion effect means the chosen replacement lowers the score. The two need not agree in magnitude or ranking because occlusion is a finite change and depends on the baseline.

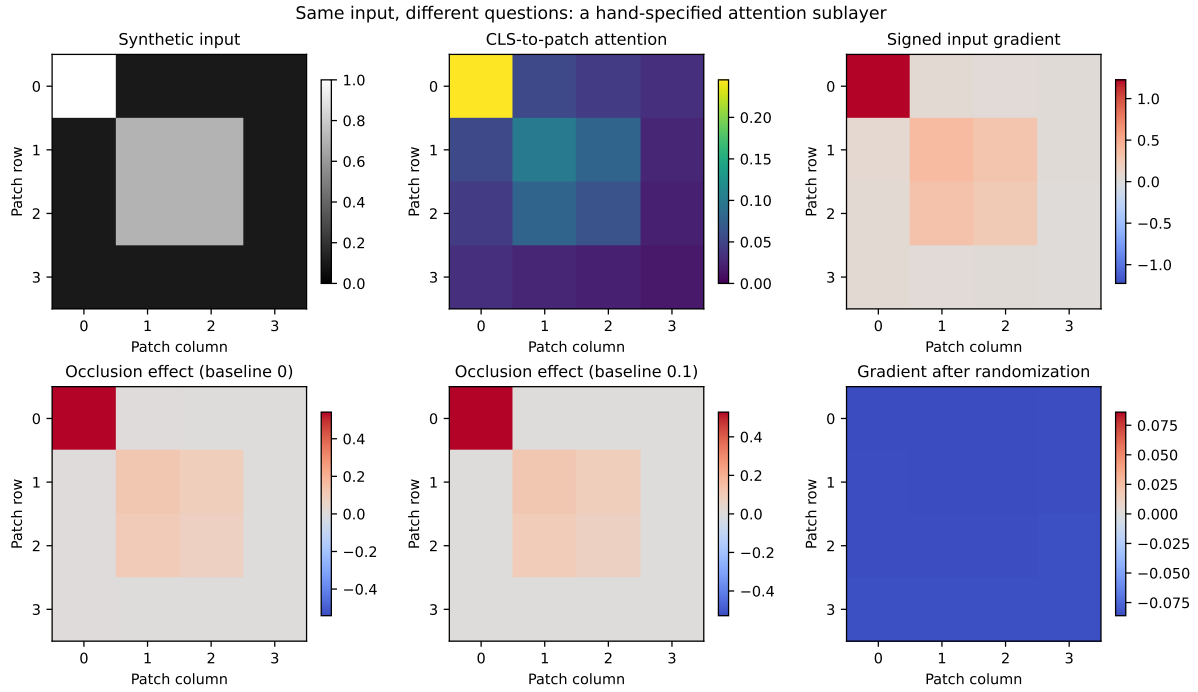



**Figure 10:** Computed maps from the same hand-specified self-attention sublayer. The central square is a synthetic pattern and the upper-left patch is a marker. Each panel has its own numerical scale: colours across panels are not directly comparable. The lower row compares two occlusion baselines and a gradient after parameter randomization. These are illustrative computational quantities, not anatomical explanations.

For the supplied input, $f = 1.586054$. Replacing the upper-left marker intensity 1.0 by the background value 0.1 leaves the central pattern unchanged and gives $f = 1.057792$, a decrease of 0.528262. This demonstrates marker sensitivity by construction; it does not demonstrate that training learned a shortcut. In a second intervention, reversing the intensity contribution to the values leaves attention weights exactly unchanged but gives $f = -1.457165$. Thus the same attention display can accompany different outputs. Replacing all parameter arrays with seeded random draws supplies a further parameter-dependence illustration. Since no model is trained, this exercise does not implement a training-label randomization test and should not be presented as reproducing Adebayo et al. (2018).

The implementation computes the gradient analytically by the chain rule and verifies every input component against central finite differences with step $10^{-6}$. A central finite difference estimates a derivative by evaluating the output just above and below an input value and dividing the difference by twice the step. The maximum absolute discrepancy in the recorded check was below $10^{-8}$. It also checks row normalization and the unchanged attention matrix in the value-only intervention. The JSON output preserves full matrices, scores, both occlusion baselines, and package versions. After setting up the project environment as described in the repository, run the example from the project root:

```
python -m brainsmodels.attention_maps
```

A notebook calls the same shared implementation in `code/notebooks/08_attention_maps.ipynb` with instructions. An extension for students is to change the occlusion baseline or readout while holding everything else fixed and predict which map will change. Applying these methods to a trained clinical system would additionally require a specified model, preprocessing, output target, valid intervention, and independent evaluation data.

**Takeaway.** A weighted output depends on scores and values; a gradient and a finite replacement describe different changes to that output.

**Check your understanding.** If both scalar values equal $-3$, what output must any normalized pair of attention weights produce?

# B Hopfield memory: a cue, an update rule, and an energy

## B.1 An explicit model of pattern completion

A stored configuration can act as an attractor: nearby starting states may settle into it. A landscape analogy is helpful if used carefully. A state is a location, the energy is its height, and the update rule restricts the available moves. Descending does not guarantee arrival at the intended valley. The construction below specifies what "descending" means, making the analogy checkable rather than explanatory by itself.

Let $s \in \{-1, +1\}^N$ be the current state: a vector of $N$ units, each with sign $-1$ or $+1$. For a single target pattern $\xi$ of the same size, define

$$w_{ij} = \frac{\xi_i \xi_j}{N} \quad (i \neq j), \qquad w_{ii} = 0, \qquad E(s) = -\frac{1}{2} \sum_{i,j} w_{ij} s_i s_j. \tag{8}$$

The weights are symmetric. The zero diagonal excludes self-input, and the factor $1/2$ avoids counting each pair twice. This is a classical associative-memory construction motivated by Hopfield (1982); the following calculations spell out the particular teaching example.

For a selected unit $i$, compute its local field $h_i = \sum_j w_{ij} s_j$, the weighted input from the other units. Set $s_i' = +1$ if $h_i > 0$, $s_i' = -1$ if $h_i < 0$, and retain $s_i$ if $h_i = 0$. The prime marks the new value. All other units remain unchanged; this is an asynchronous update. Cycle through the units, repeating until a complete pass makes no changes. The plotting implementation uses two passes for its specified cue, which is sufficient for that example.

## B.2 Why this update cannot increase energy

Only terms involving $i$ change. Symmetry and the zero diagonal give

$$\Delta E = E(s') - E(s) = -(s_i' - s_i) h_i. \tag{9}$$

If the sign stays the same, $\Delta E = 0$. If it changes, $s_i' = -s_i$ and $s_i h_i < 0$, hence $\Delta E = 2 s_i h_i < 0$. Our tie rule prevents changes at zero field. Thus every actual change strictly decreases energy. With a finite state space and repeated visits to every unit, this rule reaches a fixed point. It need not reach the global minimum or the desired pattern. Simultaneous updates change the argument and can produce cycles; the proof does not apply to arbitrary network dynamics.

### B.3 Connecting the proof to Figure 7

The target in the supplied code is

$$\xi = (1, 1, -1, -1, 1, -1, 1, -1, -1, 1, 1, -1).$$

The cue flips units 1, 4, and 8 (indices 0, 3, and 7 in Python). Write $m = \sum_i \xi_i s_i$ for the unnormalized overlap. For this single-pattern construction,

$$E(s) = -\frac{m^2 - N}{2N}. \tag{10}$$

Initially $N = 12$ and $m = 6$, so $E = -1$. After the three incorrect signs are corrected, $m$ takes values 8, 10, and 12, giving energies approximately $-2.1667$, $-3.6667$, and $-5.5$. Updates of already aligned units leave the energy unchanged. These steps explain both the drops and flat segments in the plot.

For the first correction, unit 1 is misaligned with its target: $\xi_1 = +1$ and $s_1 = -1$. Its local field is $h_1 = \xi_1(m - \xi_1 s_1)/N = 7/12$, so $s_1' = +1$. Substitution into the energy-change formula gives $\Delta E = -(1 - (-1))(7/12) = -7/6$. Adding this to the starting energy $-1$ gives $-13/6 \approx -2.1667$, matching the first drop in the recorded trajectory.

There is a revealing limitation even here: $-\xi$ has the same minimum energy as $\xi$ and is also stable. Energy minimization alone cannot distinguish the intended pattern from its inverse. With several patterns, one can use $w_{ij} = N^{-1} \sum_\mu \xi_i^\mu \xi_j^\mu$ for $i \neq j$, but interference and additional attractors complicate retrieval. A useful extension exercise is to vary the initial flips and stored patterns, recording which attractor is reached. The energy proof remains valid when its assumptions hold, while retrieval success can change. This separates a proved dynamical property from a claim about memory performance.

**Takeaway.** The energy proof concerns an update rule under stated assumptions; retrieval success additionally depends on the cue and stored patterns.

**Check your understanding.** Why does starting at $-\xi$ give a counterexample to the claim that descending energy always recovers the intended target?

## C 4E cognition: four questions about the clinical setting

The 4E vocabulary identifies related but distinct emphases (Newen et al., 2018). *Embodied* approaches attend to bodily contributions to cognition. *Embedded* approaches attend to its environmental and social setting. *Enactive* approaches emphasize activity and ongoing organism–environment engagement. *Extended* approaches ask whether some external resources can be constituents of cognition. The distinction between a causal influence and a constituent is consequential: showing that a tool changes performance is insufficient, by itself, to establish that it is part of a mind.

The following applications are proposed teaching examples, rather than empirical findings about radiology teams.

**Table 6:** Applying the four questions to the hypothetical MRI review. The examples translate concepts into investigable workflow questions.

| Emphasis | Illustrative question | A possible comparison |
|---|---|---|
| Embodied | Does the clinician's bodily state affect sustained inspection and interpretation? | Compare controlled viewing tasks across recorded fatigue or ergonomic conditions. |
| Embedded | How does access to prior scans, team discussion, and the reporting environment shape interpretation? | Compare available information and coordination under specified workflows. |
| Enactive | How does scrolling, changing the view, and revisiting a region alter the evidence obtained? | Compare active image exploration with matched prerecorded viewing. |
| Extended | Does a reliably integrated record function as part of the ongoing reasoning process? | Document patterns of reliance and disruption, while separating performance effects from the constitutive claim. |

Under suitable functional integration, the extended-mind proposal treats external resources as constituents of a cognitive process (Clark and Chalmers, 1998). A performance difference with and without a longitudinal record measures dependence on the record; identifying it as a cognitive constituent requires an additional argument.

Specify what each comparison holds constant. Prerecorded viewing can change both agency and visual evidence; withdrawing a record changes task difficulty as well as tool availability. These alternatives need controls, as in the earlier shortcut example.

The four questions broaden the explanation of a clinical decision to include the patient, clinician, tool and environment, each with a proposed observable role.

**Takeaway.** A workflow study can measure dependence on a tool while leaving the stronger claim about cognitive constitution open.

**Check your understanding.** If withdrawing a longitudinal record reduces accuracy, which task-difficulty explanation would you need to examine before making an extended-mind claim?

## References


Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity Checks for Saliency Maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://arxiv.org/abs/1810.03292.

Richard Antonello, Chandan Singh, Shailee Jain, Aliyah Hsu, Sihang Guo, Jianfeng Gao, Bin Yu, and Alexander Huth. Generative causal testing to bridge data-driven models and scientific theories in language neuroscience. arXiv:2410.00812v3, 13 June, 2026. URL https://arxiv.org/abs/2410.00812v3. Manuscript reported as accepted in Nature Neuroscience; citing the accessible version.

Emily M. Bender and Alexander Koller. Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5185–5198. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.463. URL https://doi.org/10.18653/v1/2020.acl-main.463.

Ned Block. On a confusion about a function of consciousness. *Behavioral and Brain Sciences*, 18(2):227–247, 1995. doi: 10.1017/s0140525x00038188. URL https://doi.org/10.1017/S0140525X00038188.

Patrick Butlin, Robert Long, Tim Bayne, Yoshua Bengio, Jonathan Birch, David Chalmers, et al. Identifying indicators of consciousness in AI systems. *Trends in Cognitive Sciences*, 30(6):488–501, 2026. doi: 10.1016/j.tics.2025.10.011. URL https://doi.org/10.1016/j.tics.2025.10.011.

Andy Clark and David Chalmers. The Extended Mind. *Analysis*, 58(1):7–19, 1998. doi: 10.1093/analys/58.1.7. URL https://doi.org/10.1093/analys/58.1.7.

Chris A. Cocosco, Vasken Kollokian, Remi K.-S. Kwan, and Alan C. Evans. BrainWeb: Online interface to a 3D MRI simulated brain database. *NeuroImage*, 5(4):S425, 1997. URL https://brainweb.bic.mni.mcgill.ca/brainweb/.

Cogitate Consortium, Oscar Ferrante, Urszula Gorska-Klimowska, Simon Henin, Rony Hirschhorn, Aya Khalaf, et al. Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642(8066):133–142, 2025. doi: 10.1038/s41586-025-08888-1. URL https://doi.org/10.1038/s41586-025-08888-1.

D. Louis Collins, Alex P. Zijdenbos, Vasken Kollokian, John G. Sled, Noor J. Kabani, Colin J. Holmes, and Alan C. Evans. Design and construction of a realistic digital brain phantom. *IEEE Transactions on Medical Imaging*, 17(3):463–468, 1998. doi: 10.1109/42.712135.

Javier DeFelipe. The Evolution of the Brain, the Human Nature of Cortical Circuits, and Intellectual Creativity. *Frontiers in Neuroanatomy*, 5, 2011. doi: 10.3389/fnana.2011.00029. URL https://doi.org/10.3389/fnana.2011.00029.

Adrien Doerig, Tim C. Kietzmann, Emily Allen, Yihan Wu, Thomas Naselaris, Kendrick Kay, and Ian Charest. High-level visual representations in the human brain are aligned with large language models. *Nature Machine Intelligence*, 7:1220–1234, 2025. doi: 10.1038/s42256-025-01072-0. URL https://doi.org/10.1038/s42256-025-01072-0.

FDA-NIH Biomarker Working Group. BEST (Biomarkers, EndpointS, and other Tools) Resource: Glossary. National Center for Biotechnology Information, 2016. URL https://www.ncbi.nlm.nih.gov/books/NBK338448/. Living resource; accessed 7 September 2026.

Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787. URL https://doi.org/10.1038/nrn2787.

Jack Gallifant, Majid Afshar, Saleem Ameen, Yindalon Aphinyanaphongs, Shan Chen, Giovanni Cacciamani, et al. The TRIPOD-LLM reporting guideline for studies using large language models. *Nature Medicine*, 31(1):60–69, 2025. doi: 10.1038/s41591-024-03425-5. URL https://doi.org/10.1038/s41591-024-03425-5.

Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, et al. Scaling up test-time compute with latent reasoning: A recurrent depth approach. arXiv:2502.05171, 2025. URL https://arxiv.org/abs/2502.05171.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. doi: 10.1038/s42256-020-00257-z. URL https://doi.org/10.1038/s42256-020-00257-z.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. arXiv:2412.06769, 2024. URL https://arxiv.org/abs/2412.06769.

H. Hatzikirou, D. Basanta, M. Simon, K. Schaller, and A. Deutsch. 'go or grow': the key to the emergence of invasion in tumour progression? *Mathematical Medicine and Biology*, 29(1): 49–65, 2012. doi: 10.1093/imammb/dqq011. URL https://doi.org/10.1093/imammb/dqq011.

J J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554. URL https://doi.org/10.1073/pnas.79.8.2554.

Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. doi: 10.1038/s41592-020-01008-z. URL https://doi.org/10.1038/s41592-020-01008-z.

Tomek Korbak, Mikita Balesni, Elizabeth Barnes, Yoshua Bengio, et al. Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety. arXiv:2507.11473, version 2, 2025. URL https://arxiv.org/abs/2507.11473.

Brenden M. Lake, Tomer D. Ullman, Joshua B. Tenenbaum, and Samuel J. Gershman. Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40, 2017. doi: 10.1017/s0140525x16001837. URL https://doi.org/10.1017/S0140525X16001837.

Michael A. Lepori, Kendrick Kay, and Greta Tuckute. Interpreting brain responses to language with sparse features from language models. arXiv:2606.06857, 5 June; preprint, 2026. URL https://arxiv.org/abs/2606.06857.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020. URL https://arxiv.org/abs/2005.11401.

Muxuan Liu, Ichiro Kobayashi, and Satoshi Nishida. MindAlign: Decoding inner speech from fMRI signals via multimodal embedding alignment under limited data. arXiv:2606.20696, 15 June; preprint, 2026. URL https://arxiv.org/abs/2606.20696.

Nikos K. Logothetis, Jon Pauls, Mark Augath, Torsten Trinath, and Axel Oeltermann. Neurophysiological investigation of the basis of the fMRI signal. *Nature*, 412:150–157, 2001. doi: 10.1038/35084005. URL https://doi.org/10.1038/35084005.

Alexander Selvikvåg Lundervold and Arvid Lundervold. An overview of deep learning in medical imaging focusing on MRI. *Zeitschrift für Medizinische Physik*, 29(2):102–127, 2019. doi: 10.1016/j.zemedi.2018.11.002. URL https://doi.org/10.1016/j.zemedi.2018.11.002.

Arvid Lundervold. AI-assisted glioma research: Imaging, tumor–brain interactions, and cognition. Educational code and companion presentation, 2026. URL https://github.com/arvidl/AI-assisted-glioma-research/tree/b9c1b512d93426392db6c0571f47f1b54ff05eec. Version

b9c1b512d93426392db6c0571f47f1b54ff05eec.

Arvid Lundervold, Saruar Alam, Marianne H. Hannisdal, Dorota Goplen, Martha Chekenya, and Alexander S. Lundervold. High precision segmentation of glioma and surroundings – a feasibility study using multiparametric MRI and deep learning. *Journal of Medical Engineering & Technology*, pages 1–36, 2026. doi: 10.1080/03091902.2026.2713703. URL https://doi.org/10.1080/03091902.2026.2713703. Published online 14 August 2026.

Kristina Lång, Viktoria Josefsson, Anna-Maria Larsson, Stefan Larsson, Charlotte Högberg, Hanna Sartor, et al. Artificial intelligence-supported screen reading versus standard double reading in the Mammography Screening with Artificial Intelligence trial (MASAI): a clinical safety analysis of a randomised, controlled, non-inferiority, single-blinded, screening accuracy study. *The Lancet Oncology*, 24(8):936–944, 2023. doi: 10.1016/s1470-2045(23)00298-x. URL https://doi.org/10.1016/S1470-2045(23)00298-X.

David Marr. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. The MIT Press, 2010. doi: 10.7551/mitpress/9780262514620.001.0001. URL https://doi.org/10.7551/mitpress/9780262514620.001.0001.

James L. McClelland, Bruce L. McNaughton, and Randall C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3):419–457, 1995. doi: 10.1037/0033-295x.102.3.419. URL https://doi.org/10.1037/0033-295X.102.3.419.

Kazu Nakazawa, Michael C. Quirk, Raymond A. Chitwood, Masahiko Watanabe, Mark F. Yeckel, Li Sun, et al. Requirement for hippocampal CA3 NMDA receptors in associative memory recall. *Science*, 297(5579):211–218, 2002. doi: 10.1126/science.1071795. URL https://doi.org/10.1126/science.1071795.

Albert Newen, Leon de Bruin, and Shaun Gallagher. 4E Cognition: Historical Roots, Key Concepts, and Central Issues. In Albert Newen, Leon de Bruin, and Shaun Gallagher, editors, *The Oxford Handbook of 4E Cognition*, pages 3–16. Oxford University Press, 2018. doi: 10.1093/oxfordhb/9780198735410.013.1. URL https://philosophy-cognition.com/cmc/wp-content/uploads/2019/02/Newen_deBruin_Gallagher_4E_Cognition_Historical_Roots_2018.pdf.

OpenAI. GPT-4 Technical Report. Technical Report arXiv:2303.08774, OpenAI, 2023. URL https://arxiv.org/abs/2303.08774.

Blake A. Richards, Timothy P. Lillicrap, Philippe Beaudoin, Yoshua Bengio, Rafal Bogacz, Amelia Christensen, et al. A deep learning framework for neuroscience. *Nature Neuroscience*, 22(11):1761–1770, 2019. doi: 10.1038/s41593-019-0520-2. URL https://doi.org/10.1038/s41593-019-0520-2.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241. Springer International Publishing, 2015. doi: 10.1007/978-3-319-24574-4_28. URL https://doi.org/10.1007/978-3-319-24574-4_28.

Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019. doi: 10.1038/s42256-019-0048-x. URL https://doi.org/10.1038/s42256-019-0048-x.

Anil K. Seth. Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, 17(11):565–573, 2013. doi: 10.1016/j.tics.2013.09.007. URL https://doi.org/10.1016/j.tics.2013.09.007.

Anil K. Seth and Tim Bayne. Theories of consciousness. *Nature Reviews Neuroscience*, 23(7):439–452, 2022. doi: 10.1038/s41583-022-00587-4. URL https://doi.org/10.1038/s41583-022-00587-4.

Viknesh Sounderajah, Ahmad Guni, Xiaoxuan Liu, Gary S. Collins, Alan Karthikesalingam, Sheraz R. Markar, et al. The STARD-AI reporting guideline for diagnostic accuracy stud-

ies using artificial intelligence. *Nature Medicine*, 31(10):3283–3289, 2025. doi: 10.1038/s41591-025-03953-8. URL https://doi.org/10.1038/s41591-025-03953-8.

Yannick Suter, Urspeter Knecht, Waldo Valenzuela, Michelle Notter, Ekkehard Hewer, Philippe Schucht, Roland Wiest, and Mauricio Reyes. The LUMIERE dataset: Longitudinal Glioblastoma MRI with expert RANO evaluation. *Scientific Data*, 9(1), 2022. doi: 10.1038/s41597-022-01881-7. URL https://doi.org/10.1038/s41597-022-01881-7.

Milos Suvakovic, Dom Marhoefer, Glenn Grant-Richards, and Aidan Pinero. The capacity of thought: Benchmarking Llama 3.2 in semantic fMRI neural language decoding and improving the Huth encoding-model baseline. arXiv:2607.12079v1, 13 July; preprint, 2026. URL https://arxiv.org/abs/2607.12079v1.

Kristin R. Swanson, Ellsworth C. Alvord, and James D. Murray. A quantitative model for differential motility of gliomas in grey and white matter. *Cell Proliferation*, 33(5):317–329, 2000. doi: 10.1046/j.1365-2184.2000.00177.x.

Jerry Tang, Amanda LeBel, Shailee Jain, and Alexander G. Huth. Semantic reconstruction of continuous language from non-invasive brain recordings. *Nature Neuroscience*, 26:858–866, 2023. doi: 10.1038/s41593-023-01304-9. URL https://doi.org/10.1038/s41593-023-01304-9.

Baptiste Vasey, Myura Nagendran, Bruce Campbell, David A. Clifton, Gary S. Collins, Spiros Denaxas, et al. Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *Nature Medicine*, 28(5):924–933, 2022. doi: 10.1038/s41591-022-01772-9. URL https://doi.org/10.1038/s41591-022-01772-9.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://arxiv.org/abs/1706.03762.

Patrick Y. Wen, Martin van den Bent, Gilbert Youssef, Timothy F. Cloughesy, Benjamin M. Ellingson, Michael Weller, et al. RANO 2.0: Update to the Response Assessment in Neuro-Oncology Criteria for High- and Low-Grade Gliomas in Adults. *Journal of Clinical Oncology*, 41 (33):5187–5199, 2023. doi: 10.1200/jco.23.01059. URL https://doi.org/10.1200/JCO.23.01059.

Daniel L. K. Yamins, Ha Hong, Charles F. Cadieu, Ethan A. Solomon, Darren Seibert, and James J. DiCarlo. Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the National Academy of Sciences*, 111(23):8619–8624, 2014. doi: 10.1073/pnas.1403112111. URL https://doi.org/10.1073/pnas.1403112111.

John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11):e1002683, 2018. doi: 10.1371/journal.pmed.1002683. URL https://doi.org/10.1371/journal.pmed.1002683.